

Questions

19M.2.HL.TZ1.5

The moon Phobos moves around the planet Mars in a circular orbit.

a.i. Outline the origin of the force that acts on Phobos. [1]

a.ii. Outline why this force does no work on Phobos. [1]

b.i. The orbital period T of a moon orbiting a planet of mass M is given by

$$\frac{R^3}{T^2} = kM$$

where R is the average distance between the centre of the planet and the centre of the moon.

Show that $k = \frac{G}{4\pi^2}$ [3]

b.ii.

The following data for the Mars–Phobos system and the Earth–Moon system are available:

Mass of Earth = 5.97×10^{24} kg

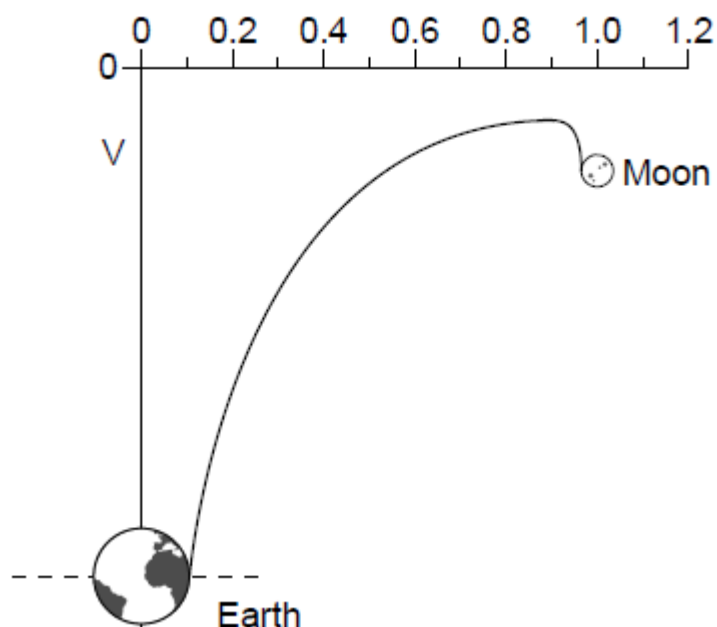
The Earth–Moon distance is 41 times the Mars–Phobos distance.

The orbital period of the Moon is 86 times the orbital period of Phobos.

Calculate, in kg, the mass of Mars. [2]

c.

The graph shows the variation of the gravitational potential between the Earth and Moon with distance from the centre of the Earth. The distance from the Earth is expressed as a fraction of the total distance between the centre of the Earth and the centre of the Moon.

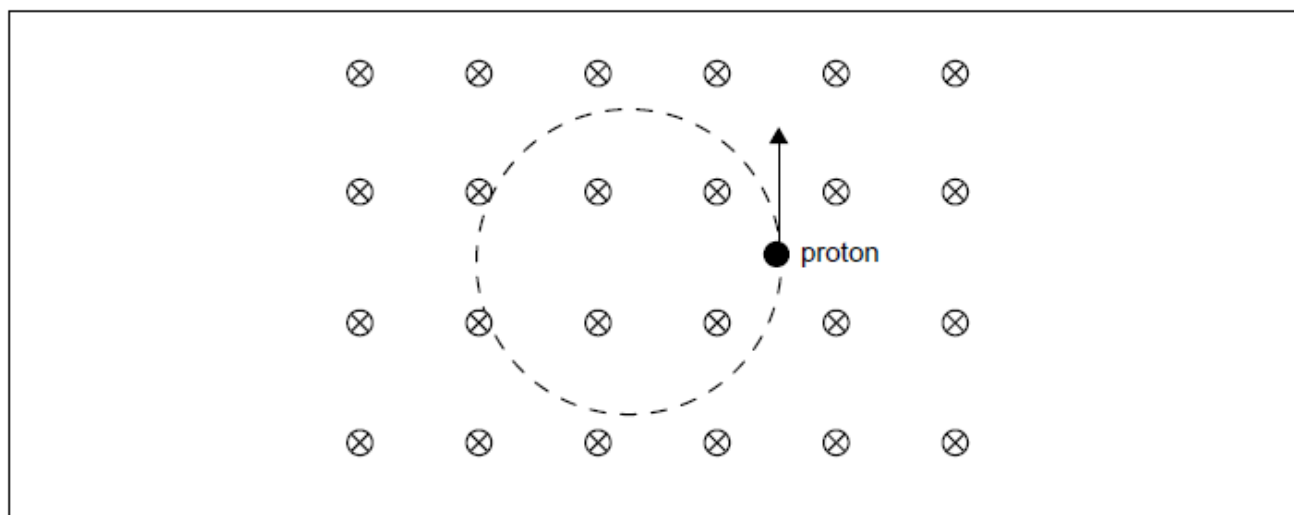


Determine, using the graph, the mass of the Moon.

[3]

19N.2.SL.TZ0.4

A proton is moving in a region of uniform magnetic field. The magnetic field is directed into the plane of the paper. The arrow shows the velocity of the proton at one instant and the dotted circle gives the path followed by the proton.

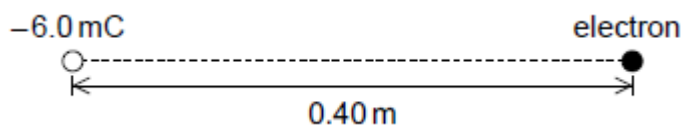


The speed of the proton is $2.0 \times 10^6 \text{ m s}^{-1}$ and the magnetic field strength B is 0.35 T .

- Explain why the path of the proton is a circle. [2]
- Show that the radius of the path is about 6 cm. [2]
 - Calculate the time for **one** complete revolution. [2]
- Explain why the kinetic energy of the proton is constant. [2]

19N.2.SL.TZ0.5

An electron is placed at a distance of 0.40 m from a fixed point charge of -6.0 mC .



a.

Show that the electric field strength due to the point charge at the position of the electron is $3.4 \times 10^8 \text{ N C}^{-1}$.

[2]

b(i). Calculate the magnitude of the initial acceleration of the electron.

[2]

b(ii). Describe the subsequent motion of the electron.

[3]

23M.2.SL.TZ1.4

(a) Show that the speed of the spacecraft is $0.80 c$ as measured in S. [1]

(i) Show that the current in Q is 0.45 A . [3]

(ii) Calculate the resistance of R. [2]

(iii) Calculate the total power dissipated in the circuit. [1]

(b)

Resistor P is removed. State and explain, without any calculations, the effect of this on the resistance of Q.

[2]

(i)

An event has coordinates $x = 0$ and $ct = 0.60 \text{ ly}$ in S. Show, using a Lorentz transformation, that the time coordinate of this event in S' is $ct' = 1.00 \text{ ly}$.

[2]

(ii)

Label, on the diagram with the letter P, the point on the ct' axis whose ct' coordinate is 1.00 ly .

[2]

(i) Draw lines to indicate R on the diagram. [2]

(ii)

Determine, using the diagram or otherwise, the space coordinate x' of event R in S' .

[2]

19N.2.SL.TZ0.7

A stationary nucleus of uranium-238 undergoes alpha decay to form thorium-234.

The following data are available.

Energy released in decay	4.27 MeV
Binding energy per nucleon for helium	7.07 MeV
Binding energy per nucleon for thorium	7.60 MeV

a.

Radioactive decay is said to be “random” and “spontaneous”. Outline what is meant by each of these terms.

Random: Spontaneous: [2]

b(i). Calculate the binding energy per nucleon for uranium-238. [3]

b(ii). Calculate the ratio $\frac{\text{kinetic energy of alpha particle}}{\text{kinetic energy of thorium nucleus}}$. [2]

23M.2.HL.TZ2.5

(a) Outline what is meant by an isotope. [1]

(a) Determine the speed of the spaceship relative to Earth. [1]

(b)

Estimate, using the spacetime diagram, the time in seconds when the flash of light reaches the spaceship according to the Earth observer.

[2]

(c)

Determine the time coordinate ct' when the flash of light reaches the spaceship, according to an observer at rest in the spaceship.

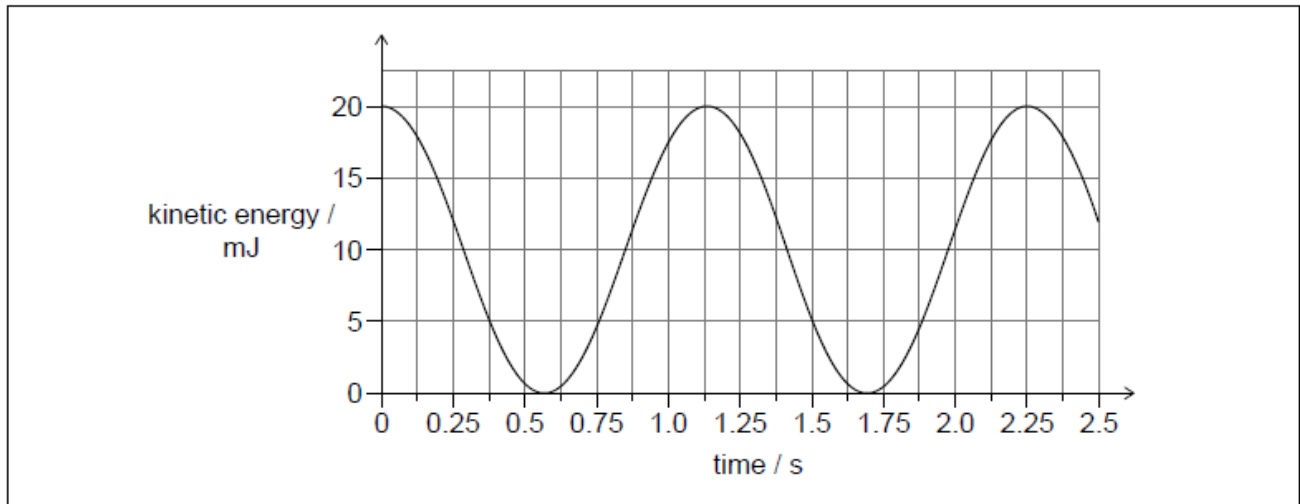
[2]

19M.2.HL.TZ1.6

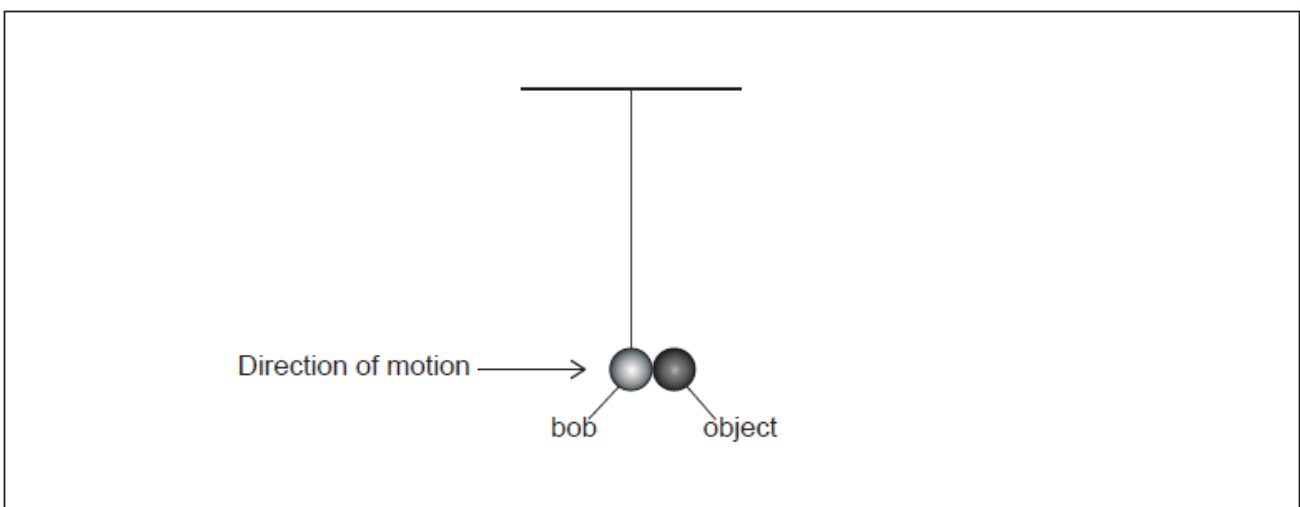
A small metal pendulum bob is suspended at rest from a fixed point with a length of thread of negligible mass. Air resistance is negligible.

The pendulum begins to oscillate. Assume that the motion of the system is simple harmonic, and in one vertical plane.

The graph shows the variation of kinetic energy of the pendulum bob with time.



When the 75 g bob is moving horizontally at 0.80 m s^{-1} , it collides with a small stationary object also of mass 75 g. The object and the bob stick together.



a.i.

Calculate, in m, the length of the thread. State your answer to an appropriate number of significant figures.

[3]

a.ii.

Label on the graph with the letter X a point where the speed of the pendulum is half that of its initial speed.

[1]

a.iii.

The mass of the pendulum bob is 75 g. Show that the maximum speed of the bob is about 0.7 m s^{-1} .

[2]

b.i.

Calculate the speed of the combined masses immediately after the collision.

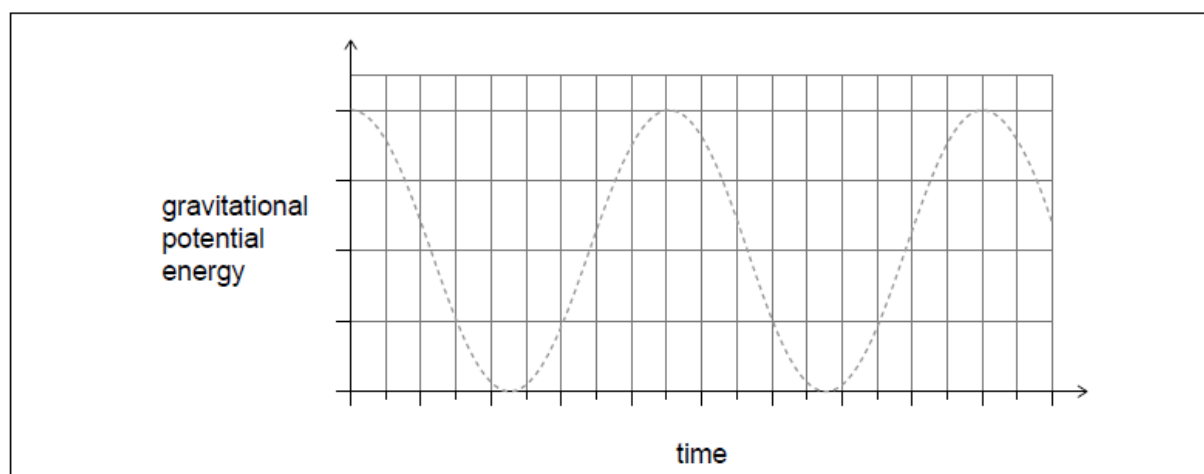
[1]

b.ii. Show that the collision is inelastic.

[3]

b.iii.

Sketch, on the axes, a graph to show the variation of gravitational potential energy with time for the bob and the object after the collision. The data from the graph used in (a) is shown as a dashed line for reference.



[2]

b.iv.

The speed after the collision of the bob and the object was measured using a sensor. This sensor emits a sound of frequency f and this sound is reflected from the moving bob. The sound is then detected by the sensor as frequency f' .

Explain why f and f' are different.

[3]

19M.2.HL.TZ2.10

A small magnet is dropped from rest above a stationary horizontal conducting ring. The south (S) pole of the magnet is upwards.

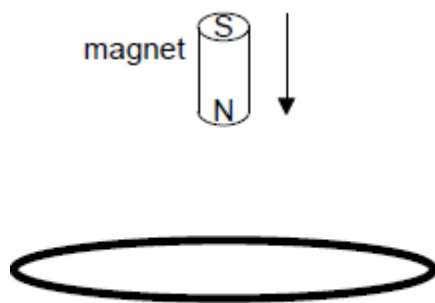


Diagram 1: side view

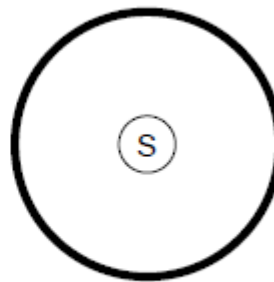


Diagram 2: view from above

a.

While the magnet is moving towards the ring, state why the magnetic flux in the ring is increasing.

[1]

b.

While the magnet is moving towards the ring, sketch, using an arrow on **Diagram 2**, the direction of the induced current in the ring.

[1]

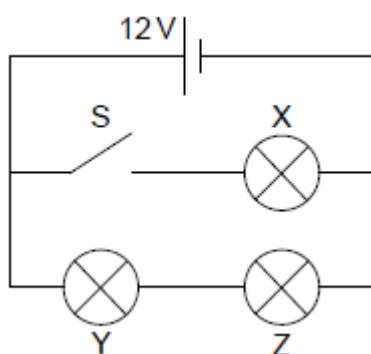
c.

While the magnet is moving towards the ring, deduce the direction of the magnetic force on the magnet.

[2]

19M.2.HL.TZ2.4

Three identical light bulbs, X, Y and Z, each of resistance $4.0\ \Omega$ are connected to a cell of emf $12\ \text{V}$. The cell has negligible internal resistance.



When fully charged the space between the plates of the capacitor is filled with a dielectric with double the permittivity of a vacuum.

a.

The switch S is initially open. Calculate the total power dissipated in the circuit.

[2]

bi.

The switch is now closed. State, without calculation, why the current in the cell will increase.

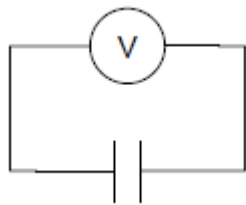
[1]

bii. The switch is now closed. Deduce the ratio $\frac{\text{power dissipated in Y with S open}}{\text{power dissipated in Y with S closed}}$.

[2]

c.

The cell is used to charge a parallel-plate capacitor in a vacuum. The fully charged capacitor is then connected to an ideal voltmeter.



The capacitance of the capacitor is $6.0 \mu\text{F}$ and the reading of the voltmeter is 12 V .

Calculate the energy stored in the capacitor.

[1]

di. Calculate the change in the energy stored in the capacitor.

[3]

dii.

Suggest, in terms of conservation of energy, the cause for the above change.

[1]

19M.2.HL.TZ2.9

A planet of mass m is in a circular orbit around a star. The gravitational potential due to the star at the position of the planet is V .

ai. Show that the total energy of the planet is given by the equation shown.

$$E = \frac{1}{2}mV \quad [2]$$

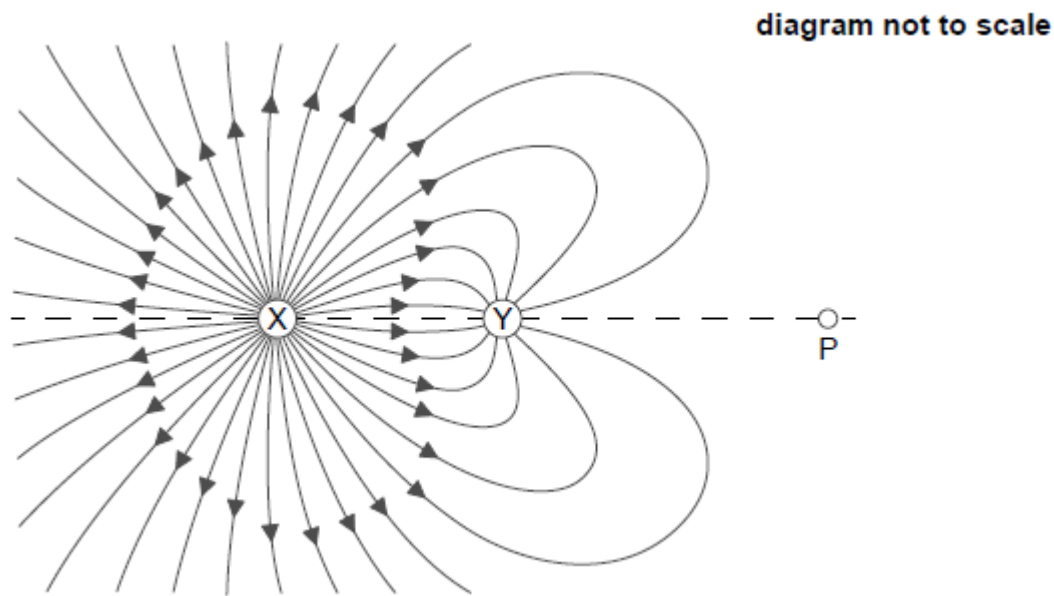
aii.

Suppose the star could contract to half its original radius without any loss of mass. Discuss the effect, if any, this has on the total energy of the planet.

[2]

b.

The diagram shows some of the electric field lines for two fixed, charged particles X and Y.



The magnitude of the charge on X is Q and that on Y is q . The distance between X and Y is 0.600 m. The distance between P and Y is 0.820 m.

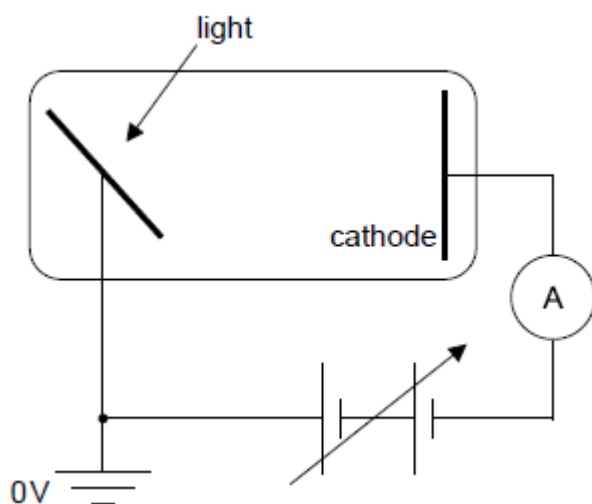
At P the electric field is zero. Determine, to **one** significant figure, the ratio $\frac{Q}{q}$.

[2]

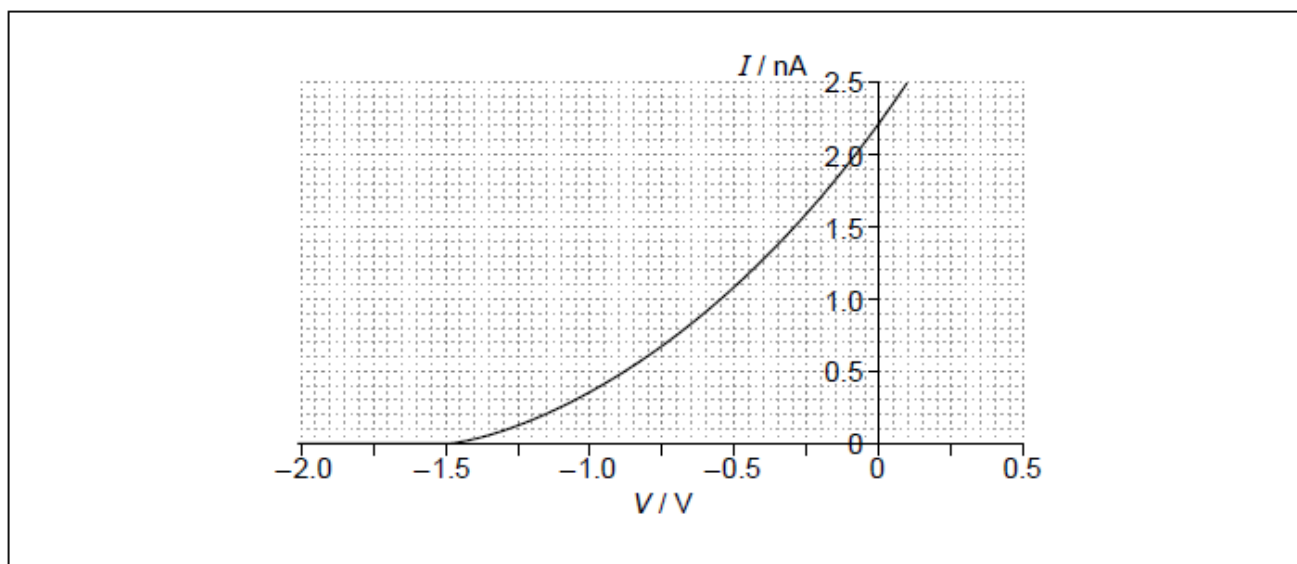
19N.2.HL.TZ0.11

Monochromatic light of very low intensity is incident on a metal surface. The light causes the emission of electrons almost instantaneously. Explain how this observation

In an experiment to demonstrate the photoelectric effect, light of wavelength 480 nm is incident on a metal surface.



The graph shows the variation of the current I in the ammeter with the potential V of the cathode.



a(i). does not support the wave nature of light. [2]

a(ii). does support the photon nature of light. [2]

b(i). Calculate, in eV, the work function of the metal surface. [3]

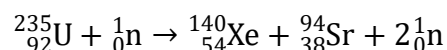
b(ii).

The intensity of the light incident on the surface is reduced by half without changing the wavelength. Draw, on the graph, the variation of the current I with potential V after this change.

[2]

20N.2.HL.TZ0.6

One possible fission reaction of uranium-235 (U-235) is



Mass of one atom of U-235 = 235 u

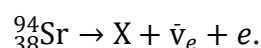
Binding energy per nucleon for U-235 = 7.59 MeV

Binding energy per nucleon for Xe-140 = 8.29 MeV

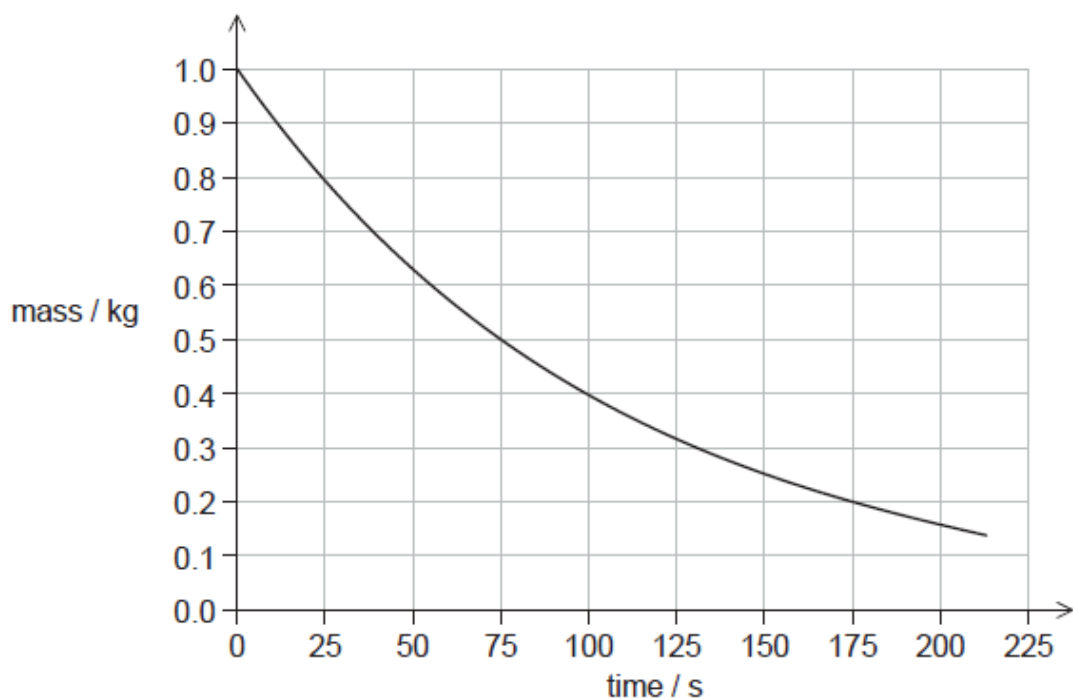
Binding energy per nucleon for Sr-94 = 8.59 MeV

A nuclear power station uses U-235 as fuel. Assume that every fission reaction of U-235 gives rise to 180 MeV of energy.

A sample of waste produced by the reactor contains 1.0 kg of strontium-94 (Sr-94). Sr-94 is radioactive and undergoes beta-minus (β^-) decay into a daughter nuclide X. The reaction for this decay is



The graph shows the variation with time of the mass of Sr-94 remaining in the sample.



a(i). State what is meant by binding energy of a nucleus. [1]

a(ii).

Outline why quantities such as atomic mass and nuclear binding energy are often expressed in non-SI units.

[1]

a(iii). Show that the energy released in the reaction is about 180 MeV. [1]

b(i). Estimate, in J kg^{-1} , the specific energy of U-235. [2]

b(ii).

The power station has a useful power output of 1.2 GW and an efficiency of 36 %. Determine the mass of U-235 that undergoes fission in one day.

[2]

b(iii).

The specific energy of fossil fuel is typically 30 MJ kg^{-1} . Suggest, with reference to your answer to (b)(i), **one** advantage of U-235 compared with fossil fuels in a power station.

[1]

c(i). Write down the proton number of nuclide X. [1]

c(ii). State the half-life of Sr-94.

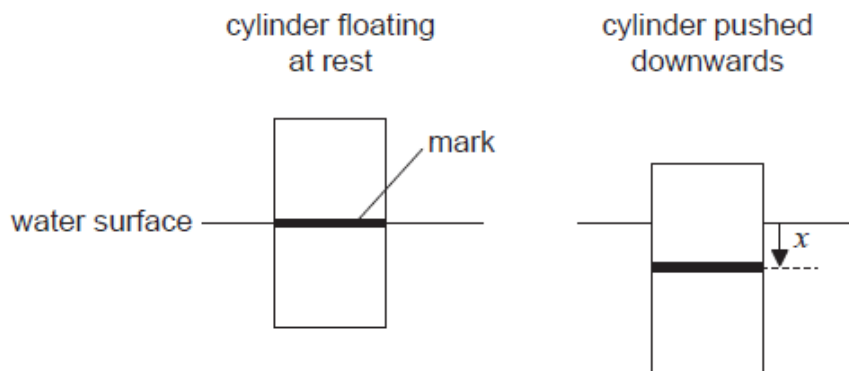
[1]

c(iii). Calculate the mass of Sr-94 remaining in the sample after 10 minutes.

[2]

20N.2.HL.TZ0.7

A vertical solid cylinder of uniform cross-sectional area A floats in water. The cylinder is partially submerged. When the cylinder floats at rest, a mark is aligned with the water surface. The cylinder is pushed vertically downwards so that the mark is a distance x below the water surface.



At time $t = 0$ the cylinder is released. The resultant vertical force F on the cylinder is related to the displacement x of the mark by

$$F = -\rho Agx$$

where ρ is the density of water.

The cylinder was initially pushed down a distance $x = 0.250 \text{ m}$.

a. Outline why the cylinder performs simple harmonic motion when released.

[1]

b.

The mass of the cylinder is 118 kg and the cross-sectional area of the cylinder is $2.29 \times 10^{-1} \text{ m}^2$. The density of water is $1.03 \times 10^3 \text{ kg m}^{-3}$. Show that the angular frequency of oscillation of the cylinder is about 4.4 rad s^{-1} .

[2]

c(i). Determine the maximum kinetic energy E_{kmax} of the cylinder.

[2]

c(ii).

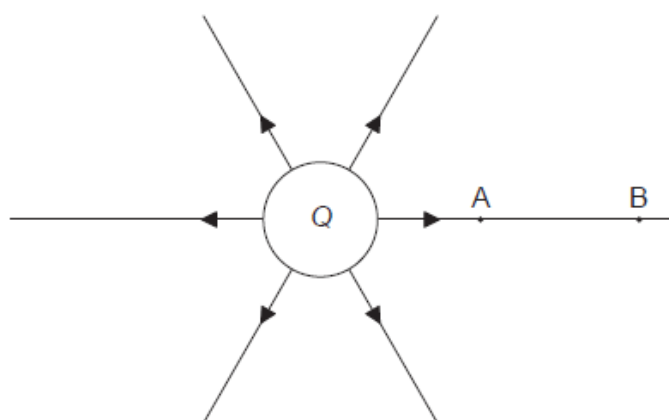
Draw, on the axes, the graph to show how the kinetic energy of the cylinder varies with time during **one** period of oscillation T .



[2]

20N.2.HL.TZ0.8

The diagram shows the electric field lines of a positively charged conducting sphere of radius R and charge Q .



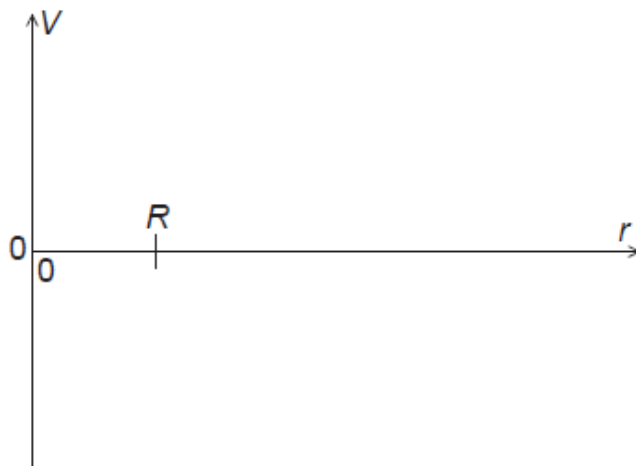
Points A and B are located on the same field line.

A proton is placed at A and released from rest. The magnitude of the work done by the electric field in moving the proton from A to B is 1.7×10^{-16} J. Point A is at a distance of 5.0×10^{-2} m from the centre of the sphere. Point B is at a distance of 1.0×10^{-1} m from the centre of the sphere.

a. Explain why the electric potential decreases from A to B. [2]

b.

Draw, on the axes, the variation of electric potential V with distance r from the centre of the sphere.



[2]

c(i). Calculate the electric potential difference between points A and B. [1]

c(ii). Determine the charge Q of the sphere. [2]

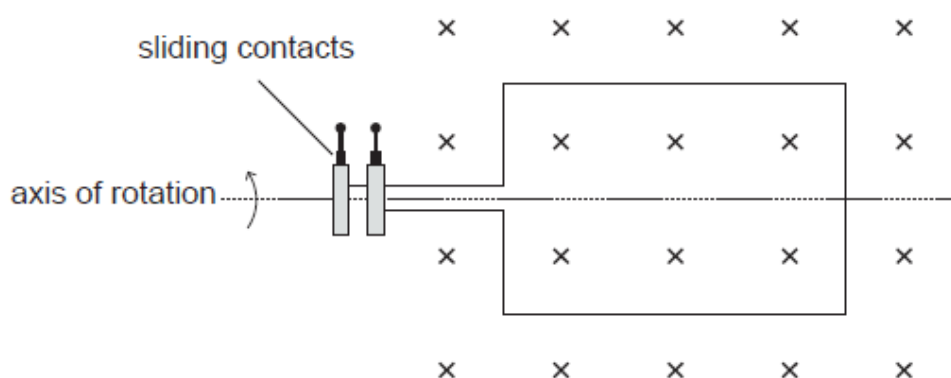
d.

The concept of potential is also used in the context of gravitational fields. Suggest why scientists developed a common terminology to describe different types of fields.

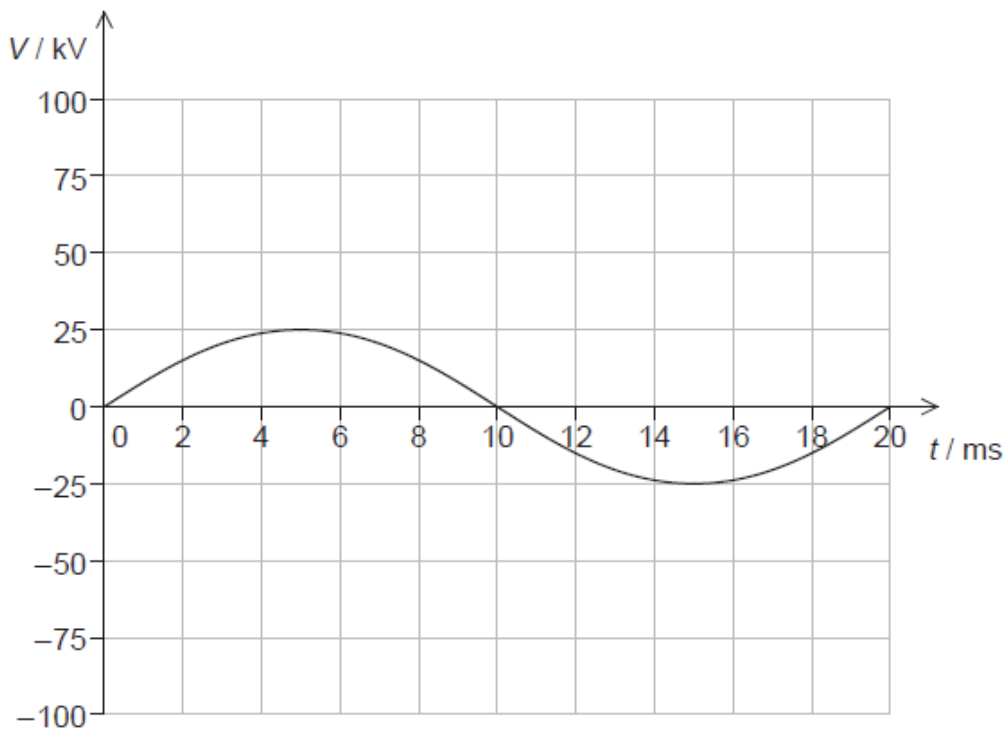
[1]

20N.2.HL.TZ0.9

The diagram shows an alternating current generator with a rectangular coil rotating at a constant frequency in a uniform magnetic field.



The graph shows how the generator output voltage V varies with time t .



Electrical power produced by the generator is delivered to a consumer some distance away.

a.

Explain, by reference to Faraday's law of induction, how an electromotive force (emf) is induced in the coil.

[3]

b(i).

The average power output of the generator is 8.5×10^5 W. Calculate the root mean square (rms) value of the generator output current.

[2]

b(ii).

The voltage output from the generator is stepped up before transmission to the consumer. Estimate the factor by which voltage has to be stepped up in order to reduce power loss in the transmission line by a factor of 2.5×10^2 .

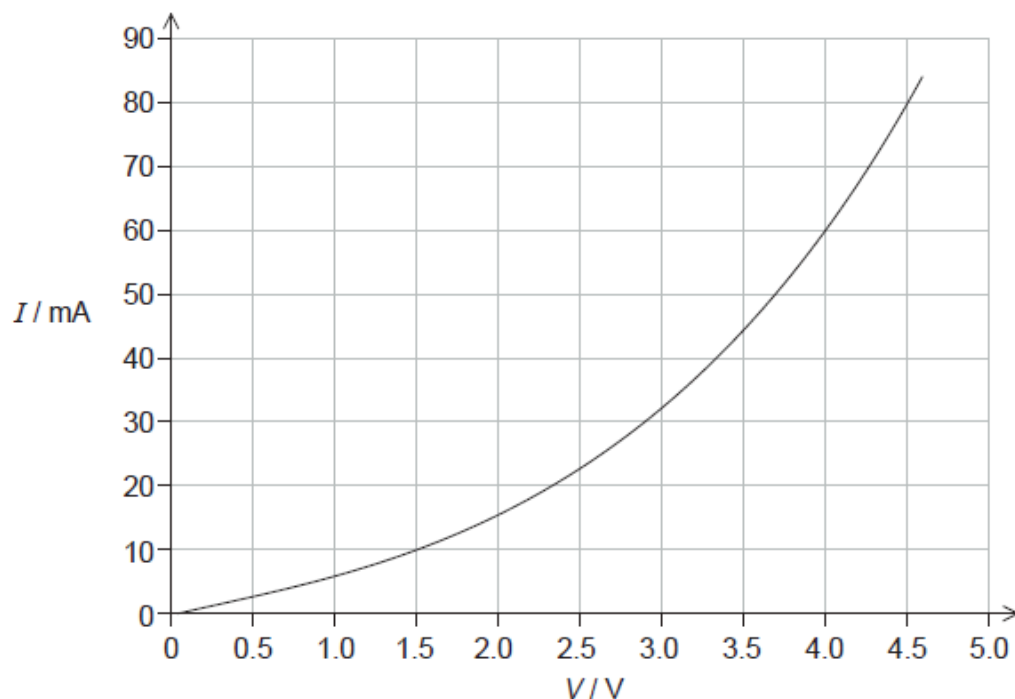
[1]

b(iii).

The frequency of the generator is doubled with no other changes being made. Draw, on the axes, the variation with time of the voltage output of the generator.

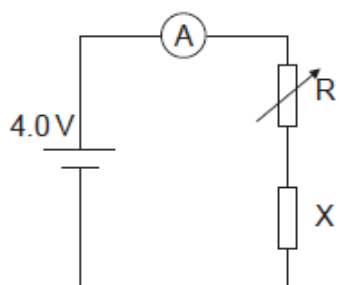
[2]

The graph shows how current I varies with potential difference V across a component X.

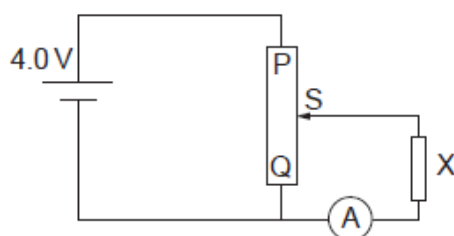


Component X and a cell of negligible internal resistance are placed in a circuit.

A variable resistor R is connected in series with component X. The ammeter reads 20 mA.



Component X and the cell are now placed in a potential divider circuit.



- a. Outline why component X is considered non-ohmic. [1]
- b(i). Determine the resistance of the variable resistor. [3]
- b(ii). Calculate the power dissipated in the circuit. [1]

c(i).

State the range of current that the ammeter can measure as the slider S of the potential divider is moved from Q to P.

[1]

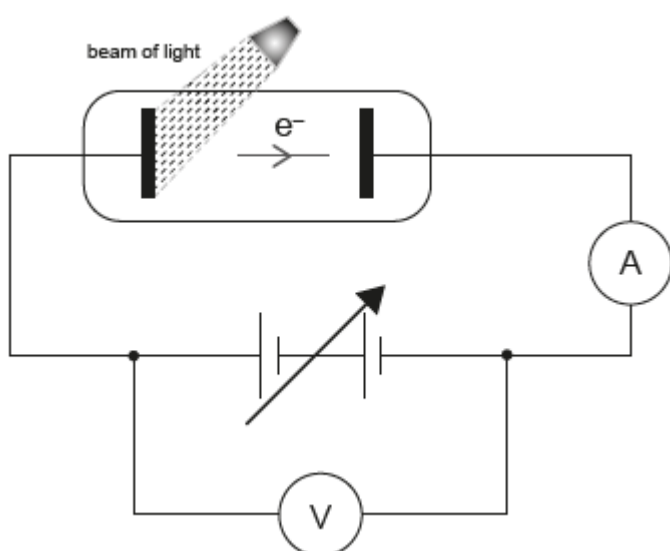
c(ii).

Describe, by reference to your answer for (c)(i), the advantage of the potential divider arrangement over the arrangement in (b).

[2]

21M.2.HL.TZ1.10

In an electric circuit used to investigate the photoelectric effect, the voltage is varied until the reading in the ammeter is zero. The stopping voltage that produces this reading is 1.40 V.



a. Describe the photoelectric effect.

[2]

b. Show that the maximum velocity of the photoelectrons is 700 km s^{-1} .

[2]

c.

The photoelectrons are emitted from a sodium surface. Sodium has a work function of 2.3 eV.

Calculate the wavelength of the radiation incident on the sodium. State an appropriate unit for your answer.

[3]

21M.2.HL.TZ2.10

The table gives data for Jupiter and three of its moons, including the radius r of each object.

Object	Mass/kg	r / m	Orbital radius around Jupiter/m
Jupiter	1.9×10^{27}	7.1×10^7	
Io	8.9×10^{22}	1.8×10^6	4.9×10^8
Ganymede	1.5×10^{23}	2.6×10^6	1.06×10^9
Callisto	1.1×10^{23}	2.4×10^6	1.88×10^9

A spacecraft is to be sent from Io to infinity.

a.

Calculate, for the surface of Io, the gravitational field strength g_{Io} due to the mass of Io. State an appropriate unit for your answer.

[2]

b.i. Show that the $\frac{\text{gravitational potential due to Jupiter at the orbit of Io}}{\text{gravitational potential due to Io at the surface of Io}}$ is about 80. [2]

b.ii.

Outline, using (b)(i), why it is not correct to use the equation $\sqrt{\frac{2G \times \text{mass of Io}}{\text{radius of Io}}}$ to calculate the speed required for the spacecraft to reach infinity from the surface of Io.

[1]

c.

An engineer needs to move a space probe of mass 3600 kg from Ganymede to Callisto. Calculate the energy required to move the probe from the orbital radius of Ganymede to the orbital radius of Callisto. Ignore the mass of the moons in your calculation.

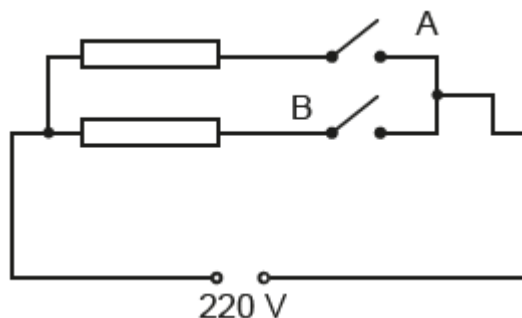
[2]

21M.2.SL.TZ1.3

A mass of 1.0 kg of water is brought to its boiling point of 100°C using an electric heater of power 1.6 kW.

A mass of 0.86 kg of water remains after it has boiled for 200 s.

The electric heater has two identical resistors connected in parallel.



The circuit transfers 1.6 kW when switch A only is closed. The external voltage is 220 V.

a.i.

The molar mass of water is 18 g mol^{-1} . Estimate the average speed of the water molecules in the vapor produced. Assume the vapor behaves as an ideal gas.

[2]

a.ii. State **one** assumption of the kinetic model of an ideal gas.

[1]

b.i.

Estimate the specific latent heat of vaporization of water. State an appropriate unit for your answer.

[2]

b.ii. Explain why the temperature of water remains at 100°C during this time.

[1]

c.

The heater is removed and a mass of 0.30 kg of pasta at -10°C is added to the boiling water.

Determine the equilibrium temperature of the pasta and water after the pasta is added. Other heat transfers are negligible.

Specific heat capacity of pasta = $1.8 \text{ kJ kg}^{-1} \text{ K}^{-1}$ [3]
 Specific heat capacity of water = $4.2 \text{ kJ kg}^{-1} \text{ K}^{-1}$

d.i. Show that each resistor has a resistance of about 30Ω .

[1]

d.ii.

Calculate the power transferred by the heater when both switches are closed.

[2]

A planet is in a circular orbit around a star. The speed of the planet is constant.

a.i.

Explain why a centripetal force is needed for the planet to be in a circular orbit.

[2]

a.ii. State the nature of this centripetal force.

[1]

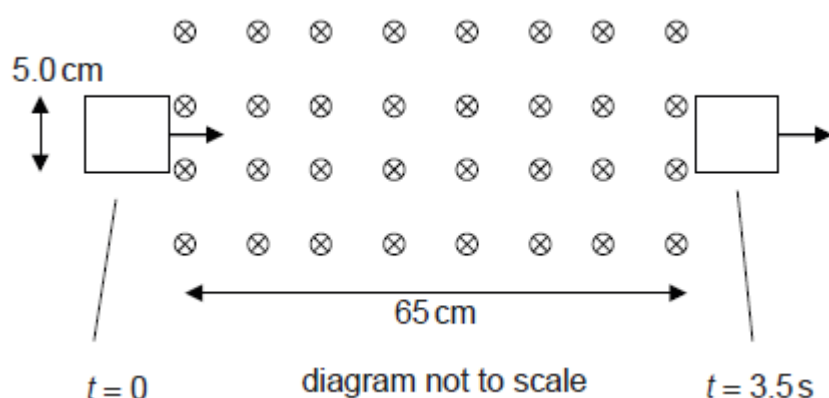
b. Determine the gravitational field of the planet.

The following data are given:

Mass of planet	$= 8.0 \times 10^{24} \text{ kg}$	[2]
Radius of the planet	$= 9.1 \times 10^6 \text{ m.}$	

21N.2.HL.TZ0.5

A square loop of side 5.0 cm enters a region of uniform magnetic field at $t = 0$. The loop exits the region of magnetic field at $t = 3.5 \text{ s}$. The magnetic field strength is 0.94 T and is directed into the plane of the paper. The magnetic field extends over a length 65 cm. The speed of the loop is constant.

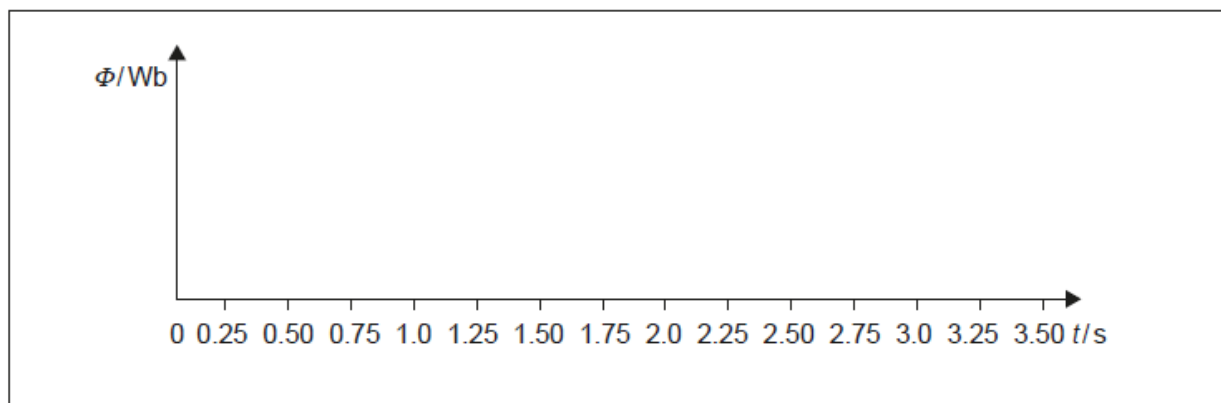


a. Show that the speed of the loop is 20 cm s^{-1} .

[1]

b.i.

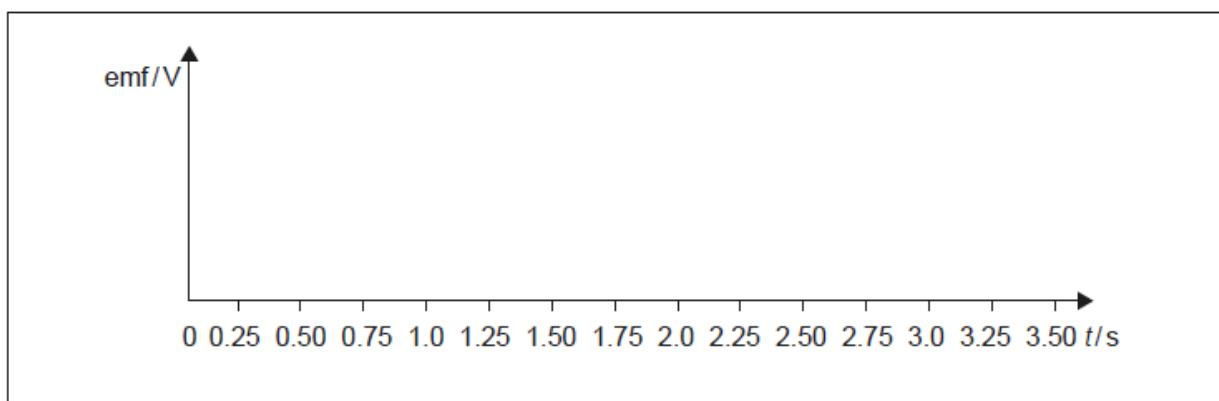
Sketch, on the axes, a graph to show the variation with time of the magnetic flux linkage Φ in the loop.



[1]

b.ii.

Sketch, on the axes, a graph to show the variation with time of the magnitude of the emf induced in the loop.



[1]

c.i.

There are 85 turns of wire in the loop. Calculate the maximum induced emf in the loop.

[2]

c.ii.

The resistance of the loop is 2.4Ω . Calculate the magnitude of the magnetic force on the loop as it enters the region of magnetic field.

[2]

d.i. Show that the energy dissipated in the loop from $t = 0$ to $t = 3.5 \text{ s}$ is 0.13 J .

[2]

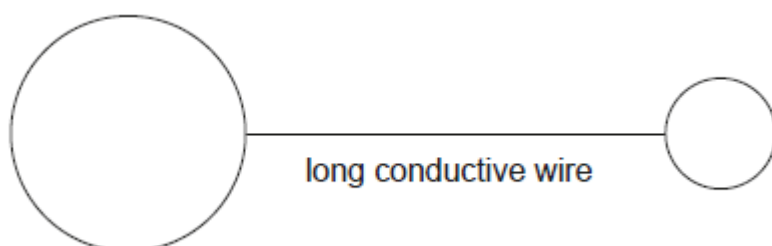
d.ii.

The mass of the wire is 18 g . The specific heat capacity of copper is $385 \text{ J kg}^{-1} \text{ K}^{-1}$. Estimate the increase in temperature of the wire.

21N.2.HL.TZ0.7

A conducting sphere has radius 48 cm. The electric potential on the surface of the sphere is $3.4 \times 10^5 \text{ V}$.

The sphere is connected by a long conducting wire to a second conducting sphere of radius 24 cm. The second sphere is initially uncharged.



a. Show that the charge on the surface of the sphere is $+18 \mu\text{C}$. [1]

b.i.

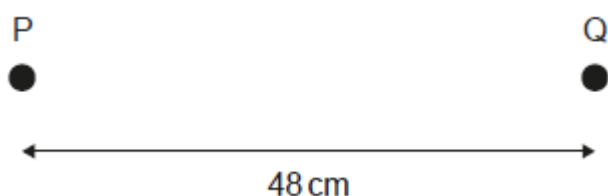
Describe, in terms of electron flow, how the smaller sphere becomes charged.

[1]

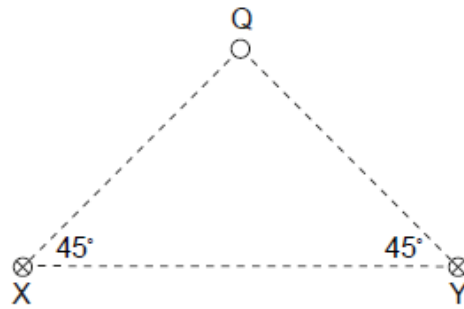
b.ii. Predict the charge on each sphere. [3]

21N.2.SL.TZ0.4

A charged particle, P, of charge $+68 \mu\text{C}$ is fixed in space. A second particle, Q, of charge $+0.25 \mu\text{C}$ is held at a distance of 48 cm from P and is then released.



The diagram shows two parallel wires X and Y that carry equal currents into the page.



Point Q is equidistant from the two wires. The magnetic field at Q due to wire X **alone** is 15 mT.

a.

The work done to move a particle of charge $0.25 \mu\text{C}$ from one point in an electric field to another is $4.5 \mu\text{J}$. Calculate the magnitude of the potential difference between the two points.

[1]

b.i. Determine the force on Q at the instant it is released.

[2]

b.ii. Describe the motion of Q after release.

[2]

c.i.

On the diagram draw an arrow to show the direction of the magnetic field at Q due to wire X **alone**.

[1]

c.ii.

Determine the magnitude and direction of the resultant magnetic field at Q. [2]

21N.2.SL.TZ0.5

Plutonium-238 (Pu) decays by alpha (α) decay into uranium (U).

The following data are available for binding energies per nucleon:

plutonium 7.568 MeV

uranium 7.600 MeV

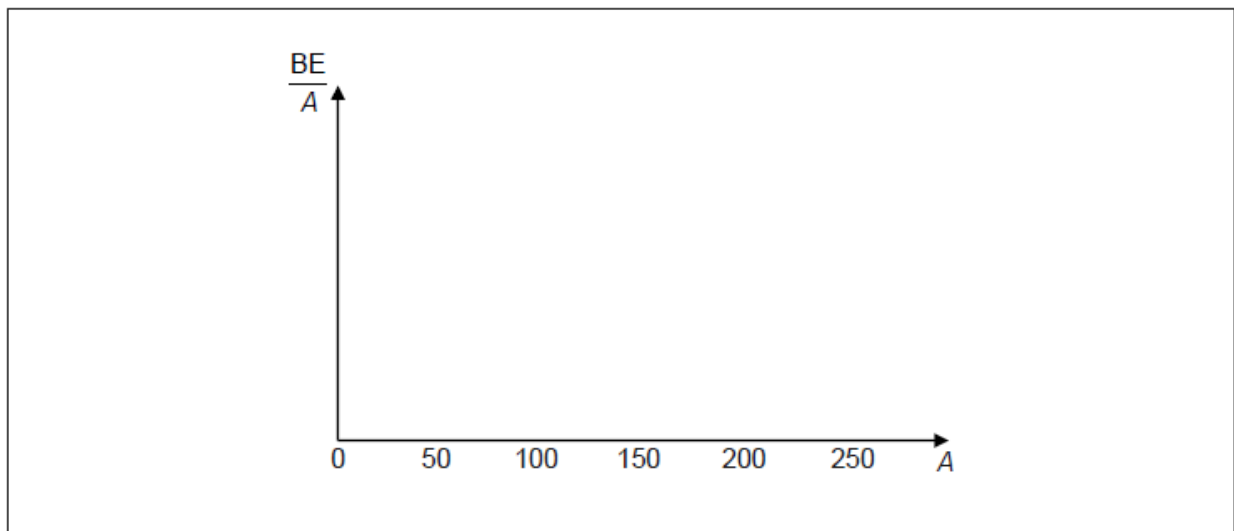
alpha particle 7.074 MeV

a.i. State what is meant by the binding energy of a nucleus.

[1]

a.ii.

Draw, on the axes, a graph to show the variation with nucleon number A of the binding energy per nucleon, $\frac{BE}{A}$. Numbers are not required on the vertical axis.



[2]

a.iii. Identify, with a cross, on the graph in (a)(ii), the region of greatest stability.

[1]

b.i. Show that the energy released in this decay is about 6 MeV.

[3]

b.ii. The plutonium nucleus is at rest when it decays.

Calculate the ratio $\frac{\text{kinetic energy of alpha particle}}{\text{kinetic energy of uranium}}$.

[2]

22N.2.HL.TZ0.6

(a)

Outline, by reference to nuclear binding energy, why the mass of a nucleus is less than the sum of the masses of its constituent nucleons.

[2]

(b.i) Calculate, in MeV, the energy released in this decay. [2]

(b.ii)

The polonium nucleus was stationary before the decay.

Show, by reference to the momentum of the particles, that the kinetic energy of the alpha particle is much greater than the kinetic energy of the lead nucleus.

[3]

(b.iii)

In the decay of polonium-210, alpha emissions can be accompanied by the emissions of gamma photons, all of the same wavelength of $1.54 \times 10^{-12} \text{ m}$. Discuss how this observation provides evidence for discrete nuclear energy levels.

[3]

(c)

A sample contains 5.0 g of pure polonium-210. The decay constant of polonium-210 is $5.8 \times 10^{-8} \text{ s}^{-1}$. Lead-206 is stable.

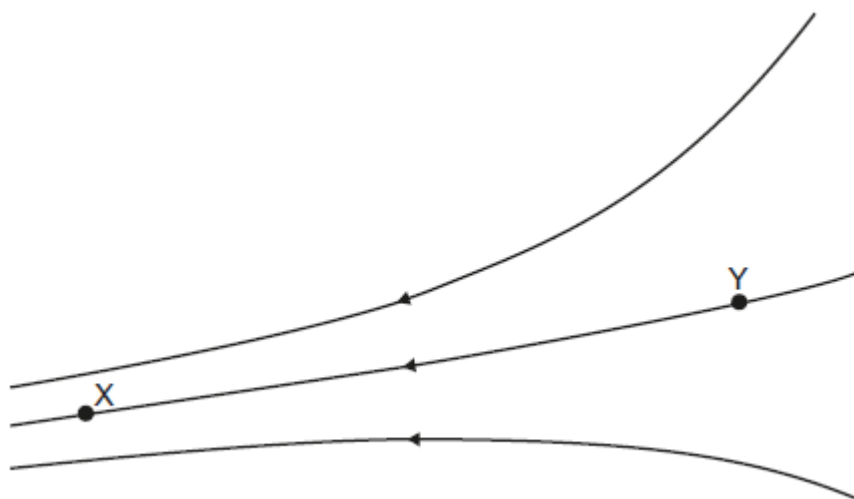
Calculate the mass of lead-206 present in the sample after one year.

[3]

22N.2.HL.TZ0.8

(a)

The diagram shows field lines for an electrostatic field. X and Y are two points on the same field line.



Outline which of the two points has the larger electric potential.

[2]

(b.i) Show that the kinetic energy of the satellite in orbit is about $2 \times 10^{10} \text{ J}$. [2]

(b.ii)

Determine the minimum energy required to launch the satellite. Ignore the original kinetic energy of the satellite due to Earth's rotation.

[2]

22N.2.SL.TZ0.5

(a) State what is meant by an ideal voltmeter. [1]

(b.i) Show that the internal resistance of the cell is about $0.7\ \Omega$. [2]

(b.ii) Show that the internal resistance of the cell is about $0.7\ \Omega$. [2]

(b.iii) Calculate the emf of the cell. [2]

(c.i)

Explain, by reference to charge carriers in the wire, how the magnetic force on the wire arises.

[2]

(c.ii) Identify the direction of the magnetic force on the wire. [1]

22N.2.SL.TZ0.6

(a)

Outline, by reference to nuclear binding energy, why the mass of a nucleus is less than the sum of the masses of its constituent nucleons.

[2]

(b.i) Calculate, in MeV, the energy released in this decay. [2]

(b.ii)

The polonium nucleus was stationary before the decay.

Show, by reference to the momentum of the particles, that the kinetic energy of the alpha particle is much greater than the kinetic energy of the lead nucleus.

[3]

(b.iii)

In the decay of polonium-210, alpha emission can be followed by the emission of a gamma photon.

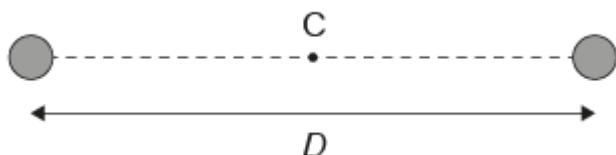
State and explain whether the alpha particle or gamma photon will cause greater ionization in the surrounding material.

[2]

23M.2.HL.TZ1.6

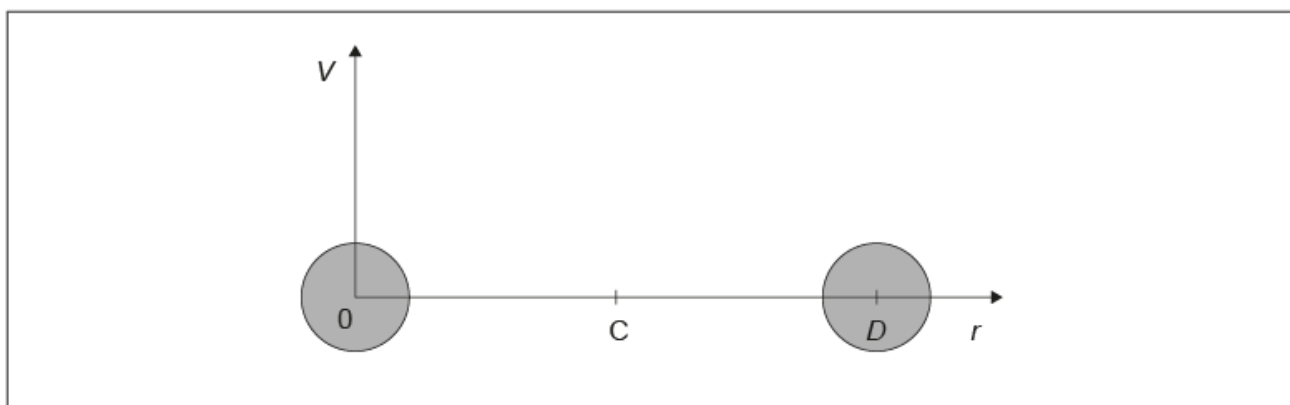
(a)

The centres of two identical fixed conducting spheres each of charge $+Q$ are separated by a distance D . C is the midpoint of the line joining the centres of the spheres.



(i)

Sketch, on the axes, how the electric potential V due to the two charges varies with the distance r from the centre of the left charge. No numbers are required. Your graph should extend from $r = 0$ to $r = D$.



[3]

(ii) Calculate the work done to bring a small charge q from infinity to point C. [2]

Data given:

$$Q = 2.0 \times 10^{-3} \text{ C},$$

$$q = 4.0 \times 10^{-9} \text{ C}$$

$$D = 1.2 \text{ m}$$

(i)

The magnitude of the net force on q is given by $\frac{32kQq}{D^3}x$. Explain why the charge q will execute simple harmonic oscillations about C.

[2]

(ii)

The mass of the charge q is 0.025 kg.

Calculate the angular frequency of the oscillations using the data in (a)(ii) and the expression in (b)(i).

[2]

(c)

The charges Q are replaced by neutral masses M and the charge q by a neutral mass m . The mass m is displaced away from C by a small distance x and released. Discuss whether the motion of m will be the same as that of q .

23M.2.HL.TZ2.9

(a)

Show, using the data, that the energy released in the decay of one magnesium-27 nucleus is about 2.62 MeV.

Mass of aluminium-27 atom = 26.98153 u

Mass of magnesium-27 atom = 26.98434 u

The unified atomic mass unit is 931.5 MeV c^{-2} .

[1]

(a) Calculate the pressure of the gas at B. [2]

(b)

Sketch, on the pV diagram, the remaining two processes BC and CA that the gas undergoes.

[2]

(i) State the conclusion that can be drawn from the existence of these two routes.

[1]

(ii)

Calculate the difference between the magnitudes of the total energy transfers in parts (a) and (b).

[1]

(iii) Explain how the difference in part (b)(ii) arises. [1]

(c) Show that the temperature of the gas at C is approximately 350°C . [2]

(i)

The smallest mass of magnesium that can be detected with this technique is $1.1 \times 10^{-8} \text{ kg}$.

Show that the smallest number of magnesium atoms that can be detected with this technique is about 10^{17} .

[2]

(ii)

A sample of glass is irradiated with neutrons so that all the magnesium atoms become magnesium-27. The sample contains 9.50×10^{15} magnesium atoms. The decay constant of magnesium-27 is $1.22 \times 10^{-3} \text{ s}^{-1}$.

Determine the number of aluminium atoms that form in 10.0 minutes after the irradiation ends.

[3]

(iii)

Estimate, in W, the average rate at which energy is transferred by the decay of magnesium-27 during the 10.0 minutes after the irradiation ends.

[2]

(d)

Explain why the change of entropy for the gas during the process BC is equal to zero.

[1]

(e)

Explain why the work done by the gas during the isothermal expansion AB is less than the work done on the gas during the adiabatic compression BC.

[1]

(f)

The quantity of trapped gas is 53.2 mol. Calculate the thermal energy removed from the gas during process CA.

[2]

23M.2.SL.TZ1.6

(a)

The moment of inertia of the rod about the axis is 0.180 kg m^2 . Show that the moment of inertia of the rod–particle system is about 0.25 kg m^2 .

[1]

(b)

Show that the angular speed of the system immediately after the collision is about 5.7 rad s^{-1} .

[2]

(c) Calculate the energy lost during the collision. [2]

(i) the angular deceleration of the rod. [1]

(ii) the number of revolutions made by the rod until it stops rotating. [2]

(e)

In another situation the rod rests on a horizontal frictionless surface with no pivot. Predict, without calculation, the motion of the rod–particle system after the collision.

[2]

23M.2.SL.TZ1.8

(a)

The moment of inertia of the rod about the axis is 0.180 kg m^2 . Show that the moment of inertia of the rod–particle system is about 0.25 kg m^2 .

[1]

(b)

Show that the angular speed of the system immediately after the collision is about 5.7 rad s^{-1} .

[2]

(c) Calculate the energy lost during the collision. [2]

(i) the angular deceleration of the rod. [1]

(ii) the number of revolutions made by the rod until it stops rotating. [2]

(e)

In another situation the rod rests on a horizontal frictionless surface with no pivot. Predict, without calculation, the motion of the rod–particle system after the collision.

[2]

23M.2.SL.TZ2.5

(a) Outline what is meant by an isotope. [1]

(i) mass number. [1]

(ii) proton number. [1]

(e)

A beta-minus particle and an alpha particle have the same initial kinetic energy. Outline why the beta-minus particle can travel further in air than the alpha particle.

[2]

EXE.2.HL.TZ0.10

(a) Calculate the angular impulse applied to the flywheel. [2]

(b)

The angular speed of the flywheel increased by 280 rad s^{-1} during the application of the angular impulse.

Determine the moment of inertia of the flywheel.

[2]

(c)

The flywheel was rotating at 150 rev per minute before the application of the angular impulse. Determine the change in angular rotational energy of the flywheel during the application of the flywheel.

[3]

EXE.2.HL.TZ0.15

(i) orbital speed; [1]

(ii) escape speed from its orbit. [1]

(i) in its initial circular orbit; [1]

(ii) in its final orbit. [1]

EXE.2.HL.TZ0.16

(a)

The radius of the dwarf planet Pluto is $1.19 \times 10^6 \text{ m}$. The acceleration due to gravity at its surface is 0.617 m s^{-2} .

Determine the escape speed for an object at the surface of Pluto.

[4]

(b)

Pluto rotates about an axis through its centre. Its rotation is in the opposite sense to that of the Earth, i.e. from east to west.

Explain the advantage of an object launching from the equator of Pluto and travelling to the west.

[3]

EXE.2.HL.TZ0.17

(a) Show that the energy of the scattered photon is about 16 keV. [1]

(b) Determine the wavelength of the incident photon. [2]

(c)

Outline why the results of the experiment are inconsistent with the wave model of electromagnetic radiation.

[2]

(d) Calculate the scattering angle of the photon. [2]

SPM.2.SL.TZ0.4

(a)

A nucleus of americium-241 has 146 neutrons. This nuclide decays to neptunium through alpha emission.

Complete the nuclear equation for this decay.



[2]

(c)

Calculate the maximum current in the chamber due to the electrons when there is no smoke in the chamber.

[3]

SPM.2.HL.TZ0.9

(a.i) Draw the electric field lines due to the charged plates. [2]

(i) Draw the electric field lines due to the charged plates. [2]

(ii)

Draw the forces acting on the oil drop, ignoring the buoyancy force.



oil drop

[2]

(iii) Show that the electric charge on the oil drop is given by [2]

$$q = \frac{\rho_o g V}{E}$$

where ρ_o is the density of oil and V is the volume of the oil drop.

Markschemes

19M.2.HL.TZ1.5

a.i.

gravitational attraction/force/field «of the planet/Mars» ✓

Do not accept "gravity".

a.ii. the force/field and the velocity/displacement are at 90° to each other **OR**

there is no change in GPE of the moon/Phobos ✓

b.i. **ALTERNATE 1** «using fundamental equations»

use of Universal gravitational force/acceleration/orbital velocity equations ✓

equating to centripetal force or acceleration. ✓ rearranges to get $k = \frac{G}{4\pi^2}$ ✓

ALTERNATE 2 «starting with $\frac{R^3}{T^2} = kM$ »

substitution of proper equation for T from orbital motion equations ✓

substitution of proper equation for M **OR** R from orbital motion equations ✓

rearranges to get $k = \frac{G}{4\pi^2}$ ✓

b.ii. $m_{\text{Mars}} = \left(\frac{R_{\text{Mars}}}{R_{\text{Earth}}}\right)^3 \left(\frac{T_{\text{Earth}}}{T_{\text{Mars}}}\right)^2 m_{\text{Earth}}$ or other consistent re-arrangement ✓

6.4×10^{23} «kg» ✓

c. read off separation at maximum potential 0.9 ✓

equating of gravitational field strength of earth and moon at that location **OR**

$$\frac{M_E}{0.9^2} = \frac{M_M}{0.1^2}$$

✓

7.4×10^{22} «kg» ✓ Allow ECF from MP1

19N.2.SL.TZ0.4

a.

magnetic force is to the left «at the instant shown»

OR

explains a rule to determine the direction of the magnetic force ✓

force is perpendicular to velocity/«direction of» motion

OR

force is constant in magnitude ✓

force is centripetal/towards the centre ✓

NOTE: Accept reference to acceleration instead of force

b(i). $qvB = \frac{mv^2}{R}$ ✓ $R = \frac{1.67 \times 10^{-27} \times 2.0 \times 10^6}{1.6 \times 10^{-19} \times 0.35}$ **OR** 0.060 « m »

NOTE: Award MP2 for full replacement or correct answer to at least 2 significant figures

b(ii). $T = \frac{2\pi R}{v}$ ✓ $T = \frac{2\pi \times 0.06}{2.0 \times 10^6} = 1.9 \times 10^{-7}$ « s » ✓

NOTE: Award [2] for bald correct answer

c. **ALTERNATIVE 1**

work done by force is change in kinetic energy ✓

work done is zero/force perpendicular to velocity ✓

NOTE: Award [2] for a reference to work done is zero hence E_k remains constant

ALTERNATIVE 2

proton moves at constant speed ✓

kinetic energy depends on speed ✓

NOTE: Accept mention of speed or velocity indistinctly in MP2

19N.2.SL.TZ0.5

a.

$$E = \frac{k \times q}{r^2} \quad \checkmark$$

$$E = \frac{8.99 \times 10^9 \times 6.0 \times 10^{-3}}{0.4^2} \quad \text{OR} \quad E = 3.37 \times 10^8 \quad \text{« N C}^{-1} \text{ » } \checkmark$$

NOTE: Ignore any negative sign.

b(i). $F = q \times E$ **OR** $F = 1.6 \times 10^{-19} \times 3.4 \times 10^8 = 5.4 \times 10^{-11}$ « N » ✓

$$a = \frac{5.4 \times 10^{-11}}{9.1 \times 10^{-31}} = 5.9 \times 10^{19} \quad \text{« m s}^{-2} \text{ » } \checkmark$$

NOTE: Ignore any negative sign.

Award [1] for a calculation leading to $a = \text{« m s}^{-2} \text{ »}$

Award [2] for bald correct answer

b(ii).

the electron moves away from the point charge/to the right «along the line joining them» ✓

decreasing acceleration ✓
increasing speed ✓

NOTE: Allow ECF from MP1 if a candidate mistakenly evaluates the force as attractive so concludes that the acceleration will increase

23M.2.SL.TZ1.4

(a)

moves 4 ly in 5 years **OR** slope of angle with time axis is 0.8 ✓

Allow evidence for mark on the graph.

(i)

Voltage across P is 1.4 «V» ✓ Voltage across Q is 4.6 «V» ✓ And $6 - 1.4 = 4.6$ «V» ✓

*Need to see a calculation involving the two voltages and the total voltage in the circuit for **MP3** (e.g. $1.4 + 4.6 = 6$).*

(ii)

Current in R is « $(0.45 - 0.4) =$ » 0.05 A ✓ So resistance is « $\frac{1.4}{0.05}$ » = 28 «Ω»

*Allow **ECF** from a(i) Allow **ECF** from **MP1***

(iii)

« 0.45×6.0 » = 2.7 «W» ✓

(b)

Q will have a smaller resistance ✓

«Because total resistance in the circuit is now larger so» the current «through the circuit/Q» is smaller / **OWTTE** ✓

*Allow similar argument for **MP2** based on voltage across Q becoming smaller.*

(i)

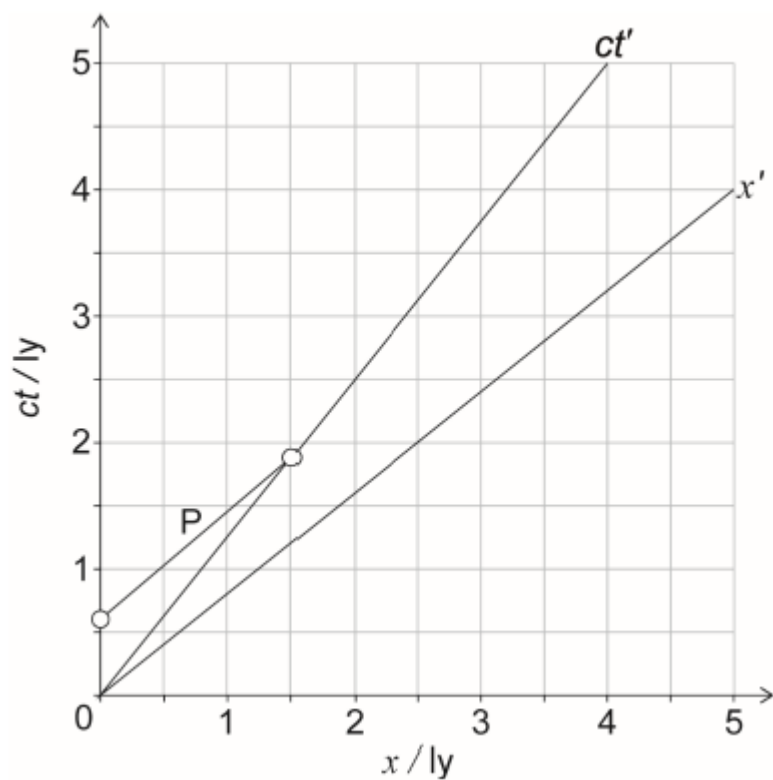
$\gamma = 1.67$ **OR** $\frac{5}{3}$ **OR** $\frac{1}{\sqrt{1 - 0.8^2}}$ ✓ $ct' = \gamma (ct - \frac{vx}{c}) = \frac{5}{3} \times (0.60 + 0)$ ✓ «= 1.00 ly»

*For **MP2**, working should be seen.*

(ii)

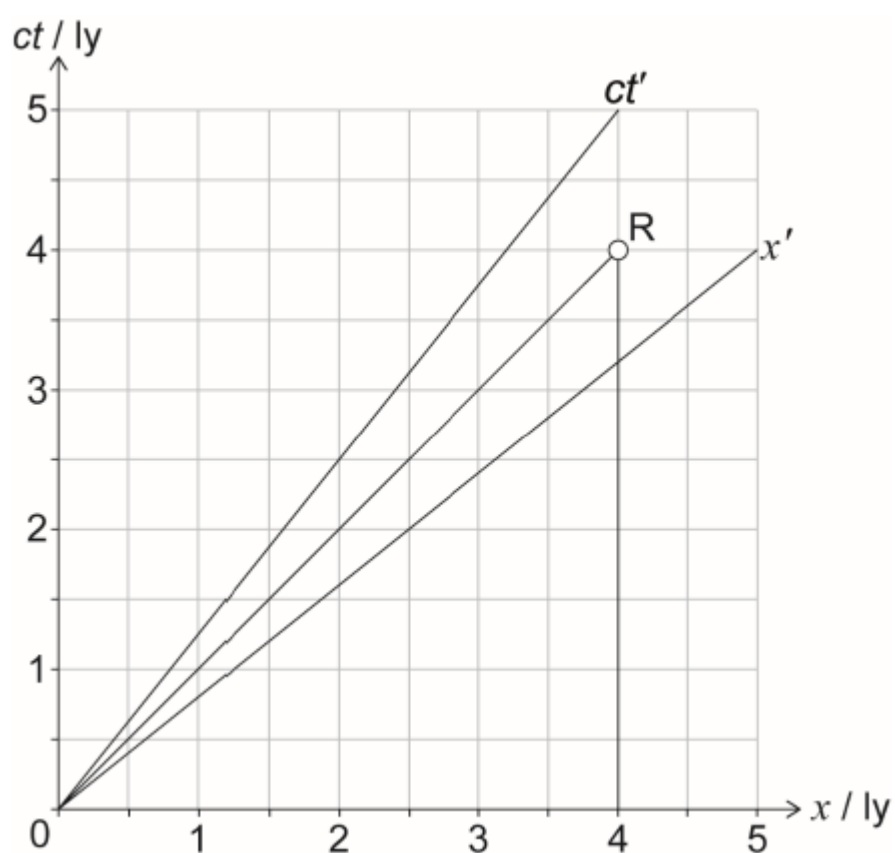
identifies point with coordinates $x = 0$, $ct = 0.60$ on vertical axis ✓

draws line parallel to the x prime axis until it intersects the prime ct axis ✓



Award [2] for correct position of P without working shown.

(i)



R located at (4,4) ✓

«as intersection of» vertical line through 4 ly and photon worldline at 45 degrees✓

Allow **MP2** even if one of the lines is not drawn.

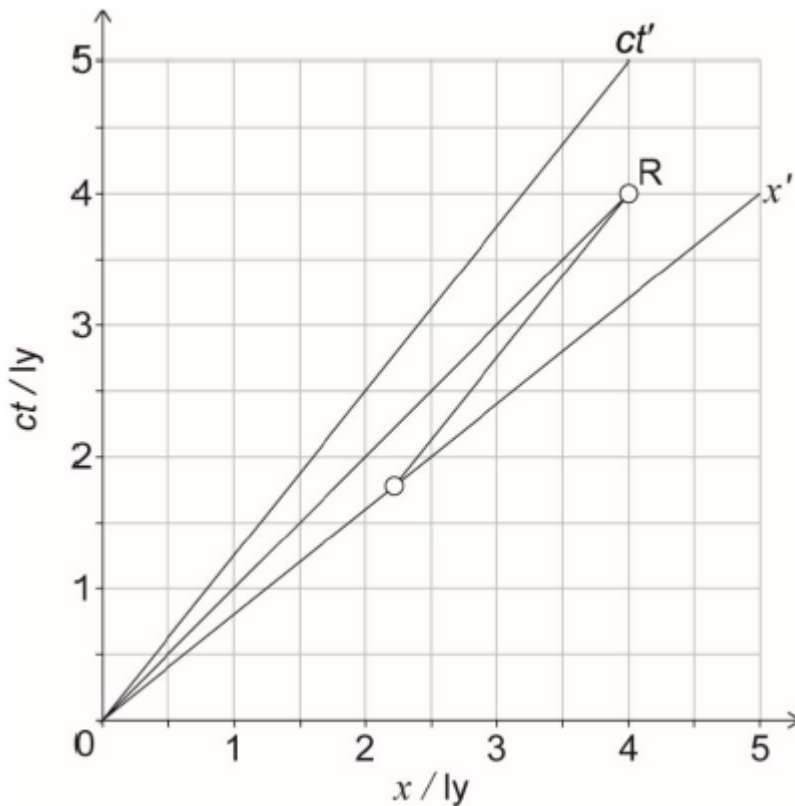
(ii)

ALTERNATIVE 1 Using diagram:

line from R parallel to prime ct axis until it intersects space axis ✓

use of scale from (b) to estimate coordinate to $x' = (1.3 \pm 0.2) \text{ ly}$ ✓

ALTERNATIVE 2



Using Lorentz transformation: event R has coordinates $x = ct = 4.00 \text{ ly}$ in S ✓

so $x' = \gamma (x - vt) = \frac{5}{3} \times (4.00 - 0.80 \times 4.00) = 1.33 \text{ ly}$ ✓

19N.2.SL.TZ0.7

a.

random:

it cannot be predicted which nucleus will decay

OR

it cannot be predicted when a nucleus will decay ✓

NOTE: OWTTE

spontaneous:

the decay cannot be influenced/modified in any way ✓

NOTE: OWTTE

b(i). 234×7.6 **OR** 4×7.07 ✓

$$BE_U = \ll 234 \times 7.6 + 4 \times 7.07 - 4.27 = \gg 1802 \ll \text{MeV} \gg \checkmark$$

$$\frac{BE_U}{A} = \ll \frac{1802}{238} = \gg 7.57 \ll \text{MeV} \gg \checkmark$$

NOTE: Allow ECF from MP2

Award **[3]** for bald correct answer

Allow conversion to J, final answer is 1.2×10^{-12}

b(ii). states or applies conservation of momentum $\checkmark$

$$\text{ratio is } \ll \frac{E_{k\alpha}}{E_{kTh}} = \frac{\frac{p^2}{2m_\alpha}}{\frac{p^2}{2m_{Th}}} = \frac{234}{4} \gg 58.5 \checkmark$$

NOTE: Award **[2]** for bald correct answer

23M.2.HL.TZ2.5

(a)

«An atom with» the same number of protons **AND** different numbers of neutrons
OR

Same chemical properties **AND** different physical properties $\checkmark$

Do **not** allow just atomic number and mass number

(a)

$$0.6c \quad \text{OR} \quad = 1.8 \times 10^8 \ll \text{m s}^{-1} \gg \checkmark$$

(b)

Line drawn at 45° from $ct = 2 \text{ km}$ to hit spaceship world line at $ct = 5 \text{ km}$ **OR**

$$ct = 1.2 / (c - 0.6c) + 2 = 5 \ll \text{km} \gg \checkmark \quad t = \frac{5000}{c} = 1.7 \times 10^{-5} \ll \text{s} \gg \checkmark$$

Award **[2]** for **BCA**

(c)

$$(ct')^2 - 0 = 5^2 - 3^2 \quad \text{OR} \quad \gamma = 1.25 \checkmark \quad ct' = 4 \ll \text{km} \gg \quad \text{OR} \quad t' = 13 \ll \mu\text{s} \gg \checkmark$$

Allow **ECF** from (b) Allow **ECF** from **MP1**

19M.2.HL.TZ1.6

a.i.

identifies T as 2.25 s $\checkmark$

$$L = 1.26 \text{ m} \checkmark$$

$$1.3 / 1.26 \ll \text{m} \gg \checkmark$$

Accept any number of s.f. for MP2.

Accept any answer with 2 or 3 s.f. for MP3.

a.ii. X labels any point on the curve where $E_k \frac{1}{4}$ of maximum/5 mJ ✓

a.iii. $\frac{1}{2}mv^2 = 20 \times 10^{-3}$ seen **OR** $\frac{1}{2} \times 7.5 \times 10^{-2} \times v^2 = 20 \times 10^{-3}$ ✓

0.73 «m s⁻¹» ✓ Must see at least 2 s.f. for MP2.

b.i. 0.40 «m s⁻¹» ✓

b.ii. initial energy 24 mJ and final energy 12 mJ ✓

energy is lost/unequal /change in energy is 12 mJ ✓

inelastic collisions occur when energy is lost ✓

b.iii. graph with same period but inverted ✓

amplitude one half of the original/two boxes throughout (by eye) ✓

b.iv. mention of Doppler effect ✓

there is a change in the wavelength of the reflected wave ✓

because the wave speed is constant, there is a change in frequency ✓

19M.2.HL.TZ2.10

a.

the magnetic field at the position of the ring is increasing «because the magnet gets closer to the ring» ✓

b. the current must be counterclockwise «in diagram 2» ✓ eg:

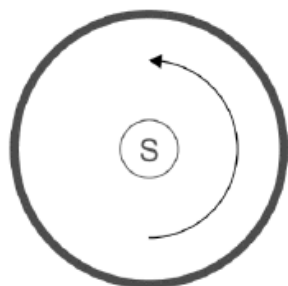


Diagram 2: view from above

c. since the induced magnetic field is upwards **OR**

by Lenz law the change «of magnetic field/flux» must be opposed **OR**

by conservation of energy the movement of the magnet must be opposed ✓

therefore the force is repulsive/upwards ✓

19M.2.HL.TZ2.4

a.

total resistance of circuit is $8.0 \, \Omega$ ✓

$$P = \frac{12^2}{8.0} = 18 \text{ W} \quad \checkmark$$

bi. «a resistor is now connected in parallel» reducing the total resistance **OR**

current through YZ unchanged and additional current flows through X ✓

bii.

evidence in calculation or statement that pd across Y/current in Y is the same as before ✓

so ratio is 1 ✓

c. $E = \frac{1}{2} CV^2 = \frac{1}{2} \times 6 \times 10^{-6} \times 12^2 = 4.3 \times 10^{-4} \text{ J} \quad \checkmark$

di. **ALTERNATIVE 1** capacitance doubles and voltage halves ✓

since $E = \frac{1}{2} CV^2$ energy halves ✓ so change is $-2.2 \times 10^{-4} \text{ J} \quad \checkmark$

ALTERNATIVE 2 $E = \frac{1}{2} CV^2$ and $Q = CV$ so $E = \frac{Q^2}{2C} \quad \checkmark$

capacitance doubles and charge unchanged so energy halves ✓

so change is $-2.2 \times 10^{-4} \text{ J} \quad \checkmark$

dii. it is the work done when inserting the dielectric into the capacitor ✓

19M.2.HL.TZ2.9

ai.

$$E = \frac{1}{2} m \frac{GM}{r} - \frac{GMm}{r} = -\frac{1}{2} \frac{GMm}{r} \quad \checkmark$$

comparison with $V = -\frac{GM}{r} \quad \checkmark$

«to give answer»

aii. **ALTERNATIVE 1**

«at the position of the planet» the potential depends only on the mass of the star / does not depend on the radius of the star ✓

the potential will not change and so the energy will not change ✓ **ALTERNATIVE 2**

r / distance between the centres of the objects / orbital radius remains unchanged ✓

since $E_{Total} = -\frac{1}{2} \frac{GMm}{r}$, energy will not change ✓

b. $\frac{kQ}{(0.600 + 0.820)^2} = \frac{kq}{0.820^2}$ ✓ $\frac{Q}{q} = \ll \frac{(0.600 + 0.820)^2}{0.820^2} = 2.9988 \approx \gg 3$ ✓

19N.2.HL.TZ0.11

a(i).

«low intensity light would» transfer energy to the electron at a low rate/slowly ✓
time would be required for the electron «to absorb the required energy» to escape/be emitted ✓

NOTE: OWTTE

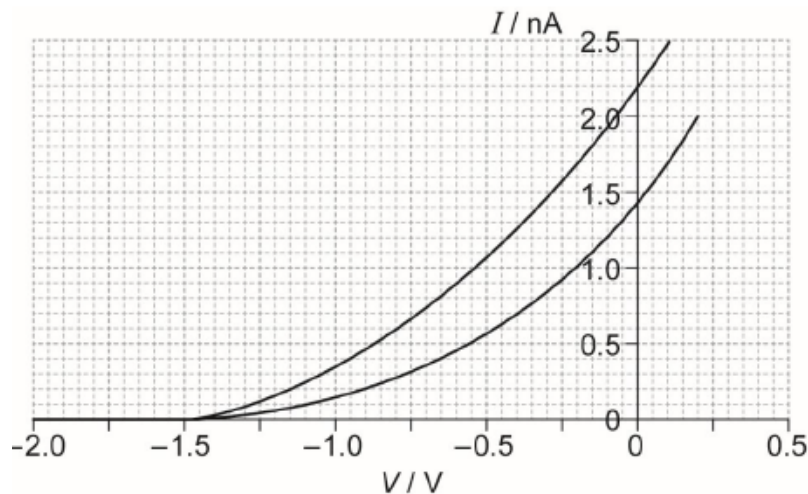
a(ii). «in the photon theory of light» the electron interacts with a single photon ✓
and absorbs all the energy **OR** and can leave the metal immediately ✓

NOTE: Reference to photon-electron collision scores MP1

b(i). $\phi = \frac{hc}{\lambda} - E_K$ ✓ $E_K = 1.5$ « eV » ✓ $\phi = \ll \frac{1.24 \times 10^{-6}}{480 \times 10^{-9}} - 1.5 = \gg 1.1$ « eV » ✓

NOTE: Allow reading from the graph of $E_K = 1.4$ leading to an answer of 1.2 «eV».

b(ii). similar curve lower than original ✓ with same horizontal intercept ✓



20N.2.HL.TZ0.6

a(i).

energy required to «completely» separate the nucleons

OR

energy released when a nucleus is formed from its constituent nucleons ✓

*Allow protons **AND** neutrons.*

a(ii). the values «in SI units» would be very small ✓

a(iii). $140 \times 8.29 + 94 \times 8.59 - 235 \times 7.59$ **OR** 184 « MeV » ✓

b(i).

see « energy = » $180 \times 10^6 \times 1.60 \times 10^{-19}$ **AND** « mass = » $235 \times 1.66 \times 10^{-27}$ ✓
 7.4×10^{13} « J kg⁻¹ » ✓

b(ii). energy produced in one day = $\frac{1.2 \times 10^9 \times 24 \times 3600}{0.36} = 2.9 \times 10^{14}$ « J » ✓

mass = $\frac{2.9 \times 10^{14}}{7.4 \times 10^{13}} = 3.9$ « kg » ✓

b(iii).

«specific energy of uranium is much greater than that of coal, hence» more energy can be produced from the same mass of fuel / per kg

OR

less fuel can be used to create the same amount of energy ✓

c(i). 39 ✓

Do not allow ${}_{39}^{94}\text{X}$ unless the proton number is indicated.

c(ii). 75 « s » ✓

c(iii). **ALTERNATIVE 1** 10 min = 8 $t_{1/2}$ ✓

mass remaining = $1.0 \times \frac{1}{2}^8 = 3.9 \times 10^{-3}$ « kg » ✓ **ALTERNATIVE 2**

decay constant = « $\frac{\ln 2}{75} = 9.24 \times 10^{-3}$ « s⁻¹ » ✓

mass remaining = $1.0 \times e^{-9.24 \times 10^{-3} \times 600} = 3.9 \times 10^{-3}$ « kg » ✓

20N.2.HL.TZ0.7

a.

the «restoring» force/acceleration is proportional to displacement ✓

Allow use of symbols i.e. $F \propto -x$ or $a \propto -x$

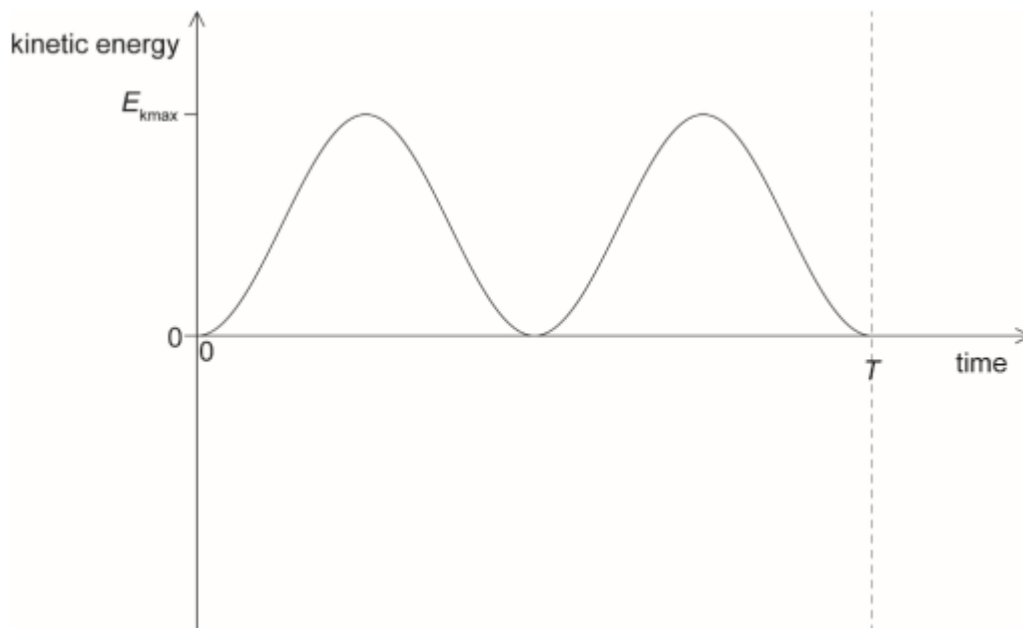
b. Evidence of equating $m\omega^2 x = \rho Agx$ «to obtain $\frac{\rho Ag}{m} = \omega^2$ » ✓

$\omega = \sqrt{\frac{1.03 \times 10^3 \times 2.29 \times 10^{-1} \times 9.81}{118}}$ **OR** 4.43 « rad s⁻¹ » ✓ *Answer to at least 3 s.f.*

c(i). « E_K is a maximum when $x = 0$ hence» $E_{K, \max} = \frac{1}{2} \times 118 \times 4.43^2 \times 0.250^2 = 0.2$ ✓

71.4 « J » ✓

c(ii). energy never negative ✓ correct shape with two maxima ✓



20N.2.HL.TZ0.8

a.

ALTERNATIVE 1

work done on moving a positive test charge in any outward direction is negative ✓
potential difference is proportional to this work «so V decreases from A to B» ✓

ALTERNATIVE 2

potential gradient is directed opposite to the field so inwards ✓
the gradient indicates the direction of increase of V «hence V increases towards the centre/decreases from A to B» ✓

ALTERNATIVE 3

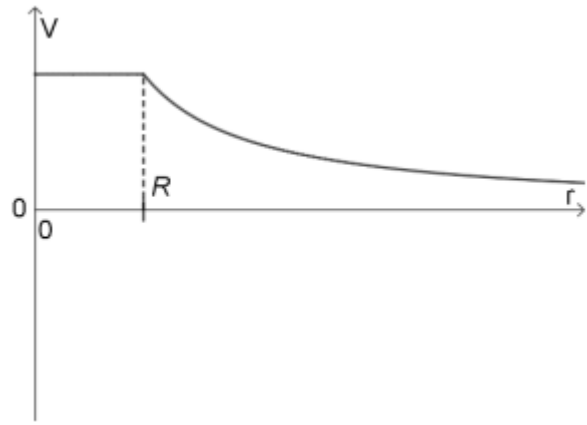
$V = \frac{kQ}{R}$ so as r increases V decreases ✓
 V is positive as Q is positive ✓

ALTERNATIVE 4

the work done per unit charge in bringing a positive charge from infinity ✓
to point B is less than point A ✓

b. curve decreasing asymptotically for $r > R$ ✓

non - zero constant between 0 and R ✓



c(i). $\ll \frac{W}{q} = \frac{1.7 \times 10^{-16}}{1.60 \times 10^{-19}} = \gg 1.1 \times 10^3 \ll V \gg \checkmark$

c(ii). $8.99 \times 10^9 \times Q \times \frac{1}{5.0 \times 10^{-2}} - \frac{1}{1.0 \times 10^{-1}} = 1.1 \times 10^3 \checkmark$
 $Q = 1.2 \times 10^{-8} \ll C \gg \checkmark$

d. to highlight similarities between «different» fields ✓

20N.2.HL.TZ0.9

a.

there is a magnetic flux «linkage» in the coil / coil cuts magnetic field ✓

this flux «linkage» changes as the angle varies/coil rotates ✓

«Faraday's law» connects induced emf with rate of change of flux «linkage» with time ✓

Do not award MP2 or 3 for answers that don't discuss flux.

b(i). $V_{\text{rms}} = \frac{25 \times 10^3}{\sqrt{2}} \ll = 17.7 \times 10^3 \ll V \gg \checkmark$ $I_{\text{rms}} = \frac{8.5 \times 10^5}{17.7 \times 10^3} = 48 \ll A \gg \checkmark$

b(ii). «power loss proportional to I^2 hence the step-up factor is $\sqrt{2.5 \times 10^2} \gg 16 \checkmark$

b(iii). peak emf doubles ✓ T halves ✓

Must show at least 1 cycle.

20N.2.SL.TZ0.5

a.

current is not «directly» proportional to the potential difference

OR

resistance of X is not constant

OR

resistance of X changes «with current/voltage» ✓

b(i). **ALTERNATIVE 1** voltage across X = 2.3 « V » ✓

voltage across R « = 4.0 - 2.3 » = 1.7 « V » ✓

resistance of variable resistor « = $\frac{1.7}{0.020}$ » = 85 « Ω » ✓ **ALTERNATIVE 2**

overall resistance « = $\frac{4.0}{0.020}$ » = 200 « Ω » ✓

resistance of X « = $\frac{2.3}{0.020}$ » = 115 « Ω » ✓

resistance of variable resistor « = 200 - 115 » = 85 « Ω » ✓

b(ii). power « = 4.0 × 0.020 » = 0.080 « W » ✓

c(i). from 0 to 60 mA ✓

c(ii). allows zero current through component X / potential divider arrangement ✓

provides greater range «of current through component X» ✓

21M.2.HL.TZ1.10

a.

electrons are ejected from the surface of a metal ✓

after gaining energy from photons/electromagnetic radiation ✓

there is a minimum «threshold» energy/frequency

OR

maximum «threshold» wavelength ✓

b. « $eV = \frac{1}{2} mv^2$ » and manipulation to get v ✓

$$v = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 1.4}{9.1 \times 10^{-31}}} \quad \text{OR} \quad 702 \text{ « kms}^{-1} \text{ » } \checkmark$$

*Must see either complete substitution or calculation to at least 3 s.f. for **MP2***

$$\text{c. } E = 2.3 + 1.4 \checkmark \quad \lambda = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{3.7 \times 1.6 \times 10^{-19}} \checkmark = 3.4 \times 10^{-7} \text{ m OR } 340 \text{ nm } \checkmark$$

*Must see an appropriate unit to award **MP3**.*

21M.2.HL.TZ2.10

a.

$$\ll \frac{GM}{r^2} = \frac{6.67 \times 10^{-11} \times 8.9 \times 10^{22}}{1.8 \times 10^6} = \gg 1.8 \checkmark$$

N kg⁻¹ **OR** m s⁻² ✓

b.i. $\frac{1.9 \times 10^{27}}{4.9 \times 10^8}$ **AND** $\frac{8.9 \times 10^{22}}{1.8 \times 10^6}$ seen ✓ « $\frac{1.9 \times 10^{27} \times 1.8 \times 10^6}{4.9 \times 10^8 \times 8.9 \times 10^{22}} = \gg 78$ ✓

For **MP1**, potentials can be seen individually or as a ratio.

b.ii.

«this is the escape speed for Io alone but» gravitational potential / field of Jupiter must be taken into account ✓

OWTTE

c. $-GM_{\text{Jupiter}} \frac{1}{1.88 \times 10^9} - \frac{1}{1.06 \times 10^9} = \ll 5.21 \times 10^7 \text{ J kg}^{-1} \gg$ ✓

« multiplies by 3600 kg to get » 1.9×10^{11} «J» ✓

Award **[2]** marks if factor of $\frac{1}{2}$ used, taking into account orbital kinetic energies, leading to a final answer of 9.4×10^{10} «J».

Allow **ECF** from **MP1** Award **[2] marks** for a bald correct answer.

21M.2.SL.TZ1.3

a.i.

$$E_k = \ll \frac{3}{2} (1.38 \times 10^{-23}) (373) \gg = 7.7 \times 10^{-21} \text{ «J» } \checkmark$$

$$v = \ll \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 6.02 \times 10^{23} \times 373}{0.018}} \gg = 720 \text{ «m s}^{-1}\text{» } \checkmark$$

a.ii. particles can be considered points «without dimensions» ✓

no intermolecular forces/no forces between particles «except during collisions» ✓

the volume of a particle is negligible compared to volume of gas ✓

collisions between particles are elastic ✓

time between particle collisions are greater than time of collision ✓

no intermolecular PE/no PE between particles ✓

Accept reference to atoms/molecules for “particle”

b.i. « $mL = Pt$ » so « $L = \frac{1600 \times 200}{0.14}$ » = 2.3×10^6 «J kg⁻¹» ✓ J kg⁻¹ ✓

b.ii.

«all» of the energy added is used to increase the «intermolecular» potential energy of the particles/break «intermolecular» bonds/**OWTTE** ✓

Accept reference to atoms/molecules for "particle"

c. use of $mc\Delta T$ ✓ $0.86 \times 4200 \times (100 - T) = 0.3 \times 1800 \times (T + 10)$ ✓

$T_{eq} = 85.69^{\circ}\text{C} \cong 86^{\circ}\text{C}$ ✓ *Accept T_{eq} in Kelvin (359 K).*

d.i. $P = \frac{v^2}{R}$ so $\frac{220^2}{1600}$ so $R = 30.25 \text{ } \Omega$ ✓

*Must see either the substituted values **OR** a value for R to at least three s.f.*

d.ii. use of parallel resistors addition so $R_{eq} = 15 \text{ } \Omega$ ✓ $P = 3200 \text{ W}$ ✓

21M.2.SL.TZ1.2

a.i.

«circular motion» involves a changing velocity ✓

«Tangential velocity» is «always» perpendicular to centripetal force/acceleration ✓

there must be a force/acceleration towards centre/star ✓

without a centripetal force the planet will move in a straight line ✓

a.ii. gravitational force/force of gravity ✓

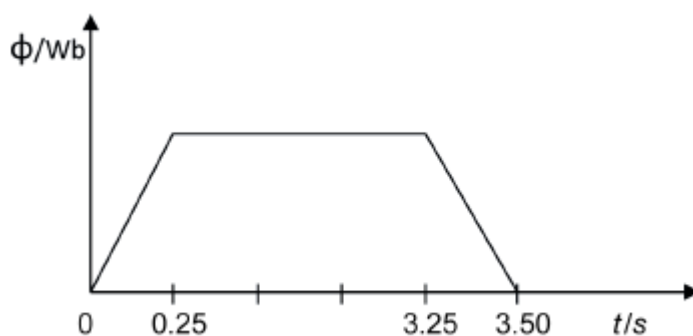
b. use of $\frac{GM}{R^2}$ ✓ 6.4 Nkg^{-1} or ms^{-2} ✓

21N.2.HL.TZ0.5

a.

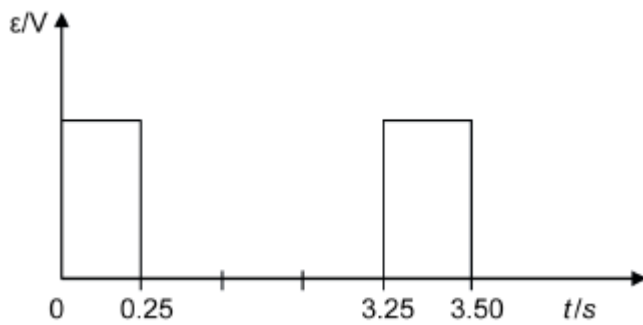
$\frac{70}{3.5}$ ✓

b.i.



shape as above ✓

b.ii.



shape as above ✓

Vertical lines not necessary to score. Allow **ECF** from **(b)(i)**.

c.i. **ALTERNATIVE 1**

maximum flux at $\ll 5.0 \times 5.0 \times 10^{-4} \times 85 \times 0.94 \gg = 0.19975 \approx 0.20 \ll \text{Wb} \gg \checkmark$

emf = $\ll \frac{0.20}{0.25} = \gg 0.80 \ll \text{V} \gg \checkmark$

ALTERNATIVE 2

emf induced in one turn = $BvL = 0.94 \times 0.20 \times 0.05 = 0.0094 \ll \text{V} \gg \checkmark$

emf = $85 \times 0.0094 = 0.80 \ll \text{V} \gg \checkmark$ Award **[2]** marks for a bald correct answer.

Allow **ECF** from **MP1**.

c.ii. $I = \ll \frac{V}{R} = \gg \frac{0.8}{2.4} \text{ OR } 0.33 \ll \text{A} \gg \checkmark$

$F = \ll NBIL = 85 \times 0.94 \times 0.33 \times 0.05 = \gg = 1.3 \ll \text{N} \gg \checkmark$ Allow **ECF** from **(c)(i)**.

Award **[2]** marks for a bald correct answer.

d.i. Energy is being dissipated for 0.50 s ✓

$$E = Fvt = 1.3 \times 0.20 \times 0.50 = \ll 0.13 \text{ J} \gg$$

OR $E = VIt = 0.80 \times 0.33 \times 0.50 = \ll 0.13 \text{ J} \gg \checkmark$ Allow **ECF** from **(b)** and **(c)**.

Watch for candidates who do not justify somehow the use of 0.5 s and just divide by 2 their answer.

d.ii. $\Delta T = \frac{0.13}{0.018 \times 385} \checkmark \quad \Delta T = 1.9 \times 10^{-2} \ll \text{K} \gg \checkmark$

Allow **[2]** marks for a bald correct answer. Award **[1]** for a **POT** error in **MP1**.

21N.2.HL.TZ0.7

a.

$$Q = \ll \frac{VR}{k} = \gg \frac{3.4 \times 10^5 \times 0.48}{8.99 \times 10^9}$$

OR

$$Q = 18.2 \ll \mu\text{C} \gg \checkmark$$

b.i. electrons leave the small sphere «making it positively charged» ✓

b.ii. $k\frac{q_1}{48} = k\frac{q_2}{24} \Rightarrow q_1 = 2q_2$ ✓ $q_1 + q_2 = 18$ ✓ so $q_1 = 12$ «μC», $q_2 = 6.0$ «μC» ✓

Award [3] marks for a bald correct answer.

21N.2.SL.TZ0.4

a.

« $V = \frac{4.5}{0.25} =$ » 18 «V» ✓

b.i. $F = \frac{8.99 \times 10^9 \times 68 \times 10^{-6} \times 0.25 \times 10^{-6}}{0.48^2}$ ✓ $F = 0.66$ «N» ✓

Award [2] marks for a bald correct answer.

*Allow symbolic k in substitutions for **MP1**.*

*Do **not** allow **ECF** from incorrect or not squared distance.*

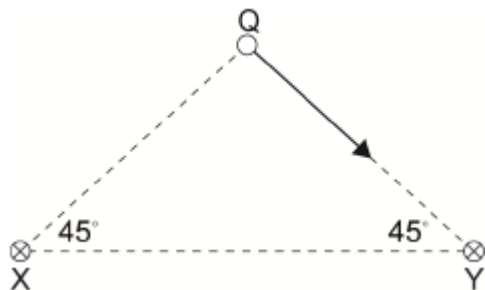
b.ii. Q moves to the right/away from P «along a straight line» **OR**

Q is repelled from P ✓

with increasing speed/Q accelerates ✓

acceleration decreases ✓

c.i.



arrow of any length as shown ✓

c.ii. «using components or Pythagoras to get» $B = 21$ «mT» ✓

directed «horizontally» to the right ✓ *If no unit seen, assume mT.*

21N.2.SL.TZ0.5

a.i.

the energy needed to «completely» separate the nucleons of a nucleus

OR

the energy released when a nucleus is assembled from its constituent nucleons ✓

Accept reference to protons **AND** neutrons.

a.ii. curve rising to a maximum between 50 and 100 ✓

curve continued and decreasing ✓ *Ignore starting point.*

Ignore maximum at alpha particle

a.iii. At a point on the peak of their graph ✓

b.i. correct mass numbers for uranium (234) and alpha (4) ✓

$234 \times 7.600 + 4 \times 7.074 - 238 \times 7.568$ «MeV» ✓ energy released 5.51 «MeV» ✓

Ignore any negative sign.

b.ii. $\ll \frac{KE_{\alpha}}{KE_U} = \gg \frac{\frac{p^2}{2m_{\alpha}}}{\frac{p^2}{2m_U}}$ **OR** $\frac{m_U}{m_{\alpha}}$ ✓ $\ll \frac{234}{4} = \gg 58.5$ ✓

Award **[2]** marks for a bald correct answer. Accept $\frac{117}{2}$ for **MP2**.

22N.2.HL.TZ0.6

(a)

according to $\Delta E = \Delta mc^2$ / identifies mass energy equivalence ✓

energy is released when nucleons come together / a nucleus is formed «so nucleus has less mass than individual nucleons»

OR

energy is required to «completely» separate the nucleons / break apart a nucleus «so individual nucleons have more mass than nucleus» ✓

Accept protons and neutrons.

(a)

according to $\Delta E = \Delta mc^2$ / identifies mass energy equivalence ✓

energy is released when nucleons come together / a nucleus is formed «so nucleus has less mass than individual nucleons»

OR

energy is required to «completely» separate the nucleons / break apart a nucleus «so individual nucleons have more mass than nucleus» ✓

Accept protons and neutrons.

(b.i)

$$(m_{\text{polonium}} - m_{\text{lead}} - m_{\alpha}) c^2 \text{ OR } (209.93676 - 205.92945 - 4.00151) \text{ OR}$$

$$\text{mass difference} = 5.8 \times 10^{-3} \checkmark$$

conversion to MeV using 931.5 to give 5.4 «MeV» ✓ *Allow ECF from MP1.*

Award [2] for a BCA. Award [1] for $8.6 \times 10^{-13} \text{ J}$.

(b.ii)

ALTERNATIVE 1

energy ratio expressed in terms of momentum, e.g. $\frac{E_{\alpha}}{E_{\text{lead}}} = \frac{p_{\alpha}^2 / 2m_{\alpha}}{p_{\text{lead}}^2 / 2m_{\text{lead}}} \checkmark$

$$p_{\alpha} = p_{\text{lead}} \text{ hence } \frac{E_{\alpha}}{E_{\text{lead}}} = \frac{m_{\text{lead}}}{m_{\alpha}} \checkmark$$

$$\frac{m_{\text{lead}}}{m_{\alpha}} \simeq \frac{206}{4} = 51.5 \Rightarrow E_{\alpha} = 51.5 E_{\text{lead}} \text{ «so } \alpha \text{ has a much greater KE» OR}$$

m_{lead} «much» greater than m_{α} «so α has a much greater KE» ✓ **ALTERNATIVE 2**

alpha particle and lead particle have equal and opposite momenta ✓

so their velocities are inversely proportional to mass ✓

but $\text{KE} \propto v^2$ «so α has a much greater KE» ✓

(b.iii)

photon energy is determined by its wavelength ✓

photons are emitted when nucleus undergoes transitions between its «nuclear» energy levels

OR photon energy equals the difference between «nuclear» energy levels ✓

OR energy is quantized / discrete ✓

photons have the same energy / a fixed value

(c)

$$\text{undecayed mass} = 5.0 \times e^{-5.8 \times 10^{-8} \times 365 \times 24 \times 60 \times 60} \text{ «} = 0.8 \text{ g» } \checkmark$$

$$\text{mass of decayed polonium «} = 5.0 - \text{undecayed mass»} = 4.2 \text{ «g» } \checkmark$$

$$\text{mass of lead «} = \frac{206}{210} \times 4.2 \text{ »} = 4.1 \text{ «g» } \checkmark$$

Allow [2] max for answers that ignore mass difference between Pb and Po (4.2 g).

Allow calculations in number of particles or moles for MP1 and MP2.

Allow ECF from MP1 and MP2.

22N.2.HL.TZ0.8

(a)

potential greater at Y ✓

«from $E = -\frac{\Delta V_e}{\Delta r}$ » the potential increases in the direction opposite to field strength
 «so from X to Y»

OR opposite to the direction of the field lines, «so from X to Y» **OR**

«from $W = q \Delta V_e$ » work done to move a positive charge from X to Y is positive «so the potential increases from X to Y» ✓

(a)

potential greater at Y ✓

«from $E = -\frac{\Delta V_e}{\Delta r}$ » the potential increases in the direction opposite to field strength
 «so from X to Y»

OR opposite to the direction of the field lines, «so from X to Y» **OR**

«from $W = q \Delta V_e$ » work done to move a positive charge from X to Y is positive «so the potential increases from X to Y» ✓

(b.i)

orbital radius = $6.4 \times 10^6 + 5.0 \times 10^5$ « = 6.9×10^6 m » ✓

$KE = \frac{1}{2} \times 8.0 \times 10^2 \times \frac{6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{6.9 \times 10^6}$ **OR** 2.3×10^{10} «J» ✓

Award [1] max for answers ignoring orbital height ($KE = 2.5 \times 10^{10}$ J).

(b.ii)

change in PE = $6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times 8.0 \times 10^2 \frac{1}{6.4 \times 10^6} - \frac{1}{6.9 \times 10^6} =$ «
 3.6×10^9 J » ✓

energy needed = $KE + \Delta PE = 2.7 \times 10^{10}$ «J» ✓ *Allow ECF from 8(b)(i).*

22N.2.SL.TZ0.5

(a)

infinite resistance **OR** no current is flowing through it ✓

(b.i)

current = $\frac{1.47}{50.0} = 2.94 \times 10^{-2}$ «A» ✓ $r = \frac{1.49}{2.94 \times 10^{-2}} - 50.0$ **OR** 0.68 «Ω» ✓

For **MP2**, allow any other correctly substituted expression for r .

(b.ii)

$$29.4(50.0 + r) = 139(10.0 + r) \checkmark$$

OR

attempt to solve for r , e.g. $29.4 \times 50.0 - 139 \times 10.0 = r(139 - 29.4)$

$0.73 \llcorner \Omega \llcorner \checkmark$ Do **not** allow working backwards from 0.7Ω .

(b.iii)

$$139 \times 10^{-3}(10.0 + 0.73) \quad \textbf{OR} \quad 29.4 \times 10^{-3}(50.0 + 0.73) \checkmark$$

$1.49 \llcorner V \llcorner \checkmark$

Watch for **ECF** from 5(b)(i).

(c.i)

charge/carriers are moving in a magnetic field $\checkmark$

there is a magnetic force on them / quote $F = qvB$ **OR**

this creates a magnetic field that interacts with the external magnetic field $\checkmark$

Accept *electrons*.

For **MP2**, the force must be identified as acting on charge / carriers.

(c.ii)

into the plane «of the paper» $\checkmark$

22N.2.SL.TZ0.6

(a)

according to $\Delta E = \Delta mc^2$ / identifies mass energy equivalence $\checkmark$

energy is released when nucleons come together / a nucleus is formed «so nucleus has less mass than individual nucleons»

OR

energy is required to «completely» separate the nucleons / break apart a nucleus «so individual nucleons have more mass than nucleus» $\checkmark$

Accept *protons and neutrons*.

(a)

according to $\Delta E = \Delta mc^2$ / identifies mass energy equivalence ✓

energy is released when nucleons come together / a nucleus is formed «so nucleus has less mass than individual nucleons»

OR

energy is required to «completely» separate the nucleons / break apart a nucleus «so individual nucleons have more mass than nucleus» ✓

Accept protons and neutrons.

(b.i)

$(m_{\text{polonium}} - m_{\text{lead}} - m_{\alpha}) c^2$ **OR** $(209.93676 - 205.92945 - 4.00151)$ **OR**

mass difference = 5.8×10^{-3} ✓

conversion to MeV using 931.5 to give 5.4 «MeV» ✓ *Allow ECF from MP1.*

Award [2] for a BCA. Award [1] for 8.6×10^{-13} J.

(b.ii)

ALTERNATIVE 1

energy ratio expressed in terms of momentum, e.g. $\frac{E_{\alpha}}{E_{\text{lead}}} = \frac{p_{\alpha}^2 / 2m_{\alpha}}{p_{\text{lead}}^2 / 2m_{\text{lead}}}$ ✓

$p_{\alpha} = p_{\text{lead}}$ hence $\frac{E_{\alpha}}{E_{\text{lead}}} = \frac{m_{\text{lead}}}{m_{\alpha}}$ ✓

$\frac{m_{\text{lead}}}{m_{\alpha}} \simeq \frac{206}{4} = 51.5 \Rightarrow E_{\alpha} = 51.5 E_{\text{lead}}$ «so α has a much greater KE» **OR**

m_{lead} «much» greater than m_{α} «so α has a much greater KE» ✓ **ALTERNATIVE 2**

alpha particle and lead particle have equal and opposite momenta ✓

so their velocities are inversely proportional to mass ✓

but $KE \propto v^2$ «so α has a much greater KE» ✓

(b.iii)

alpha particle ✓

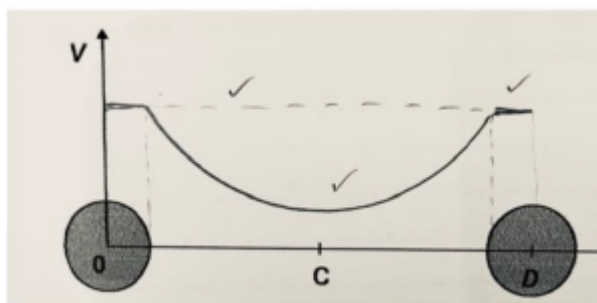
is electrically charged hence more likely to interact with electrons «in the surrounding material» ✓

23M.2.HL.TZ1.6

(a)

Constant, non-zero within spheres ✓ A clear, non-zero positive minimum at C ✓

Symmetric bowl shaped up curved shape in between ✓

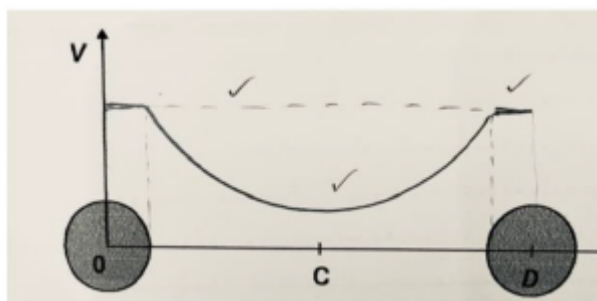


Do **not** allow a bowl shaped down curve for **MP3**.

(i)

Constant, non-zero within spheres ✓ A clear, non-zero positive minimum at C ✓

Symmetric bowl shaped up curved shape in between ✓



Do **not** allow a bowl shaped down curve for **MP3**.

(ii)

$$V \ll 2 \times \frac{8.99 \times 10^9 \times 2.0 \times 10^{-3}}{0.60} \gg = 6.0 \times 10^7 \ll V \gg \checkmark$$

$$W = \ll qV = 6.0 \times 10^7 \times 4.0 \times 10^{-9} \gg = 0.24 \ll J \gg \checkmark \quad \text{Allow ECF from MP1}$$

(i)

The restoring force/acceleration is opposite to the displacement/towards equilibrium / **OWTTE** ✓

and proportional to displacement from equilibrium / **OWTTE** ✓

Allow discussions based on the diagram (such as towards C for towards equilibrium).

Accept $F \propto x$ **OR** $a \propto x$ for **MP2**

(ii)

$$\omega = \sqrt{\frac{32kQq}{mb^3}} \quad \text{OR use of } F = m\omega^2 r \quad \text{OR } F = 1.33 \times \quad \text{OR } a = 53.3 \times \checkmark$$

$$\ll \sqrt{\frac{32 \times 8.99 \times 10^9 \times 2.0 \times 10^{-3} \times 4.0 \times 10^{-9}}{0.025 \times 1.2^3}} \gg = 7.299 \ll s^{-1} \gg$$

(c)

the net force will no longer be a restoring force/directed towards equilibrium **OR**

the gravitational force is attractive/neutral mass would be pulled towards larger masses/ **OWTTE** ✓

«and so» no, motion will not be the same/no longer be SHM / **OWTTE** ✓

23M.2.HL.TZ2.9

(a)

$(26.98434 - 26.98153) \times 931.5$ **OR** 2.6175 «MeV» seen ✓

(a)

use of $pV = \text{constant}$ ✓ $P_B = 43$ «Kpa» ✓ **Award [2] for BCA**

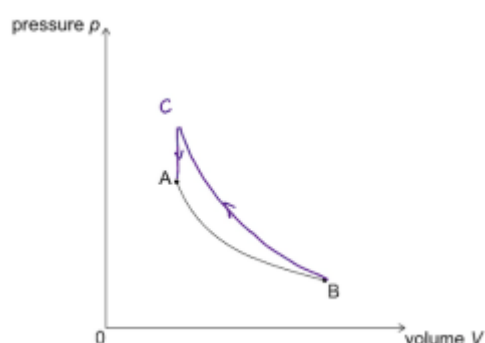
(b)

concave curved line from B to locate C with a higher pressure than A ✓

vertical line joining C to A ✓

Allow **ECF** from **MP1** i.e., **award [1]** for first process locating C at a lower pressure than A, then vertical line to A.

Arrows on the processes are not needed. Point C need not be labelled.



(i)

evidence for nuclear energy levels ✓

(ii)

Difference = $2.6175 - (1.76656 + 0.84376) = 2.6175 - 2.61032 = 0.007195$ «MeV» **OR**

Difference = $2.6175 - (1.59587 + 1.01445) = 2.6175 - 2.61032 = 0.007195$ «MeV» ✓

(iii)

Another particle/«anti» neutrino is emitted «that accounts for this mass / energy» ✓

(c)

ALTERNATIVE 1 use of $TV^{\frac{2}{3}} = \text{constant}$ «so $300 (3V_A)^{\frac{2}{3}} = T_C (V_A)^{\frac{2}{3}}$ » ✓

$T_C = 624$ «K» **OR** $T_C = 351$ «°C» ✓ **ALTERNATIVE 2**

use of $pV^{\frac{5}{3}}$ to get either $p_C = 43 (3)^{\frac{5}{3}}$ **OR** $p_C = 268$ «kPa» ✓

« $T_C = 268 \times 300/129 = \text{so}$ » $T_C = 624$ «K» **OR** $T_C = 351$ «°C» ✓

(i)

So $1.1 \times 10^{-8} \text{ kg} \equiv \frac{1.1}{0.027} \times 10^{-8}$ «mol» **OR**

Mass of atom = $27 \times 1.66 \times 10^{-27}$ «kg» ✓ $2.4 - 2.5 \times 10^{17}$ atoms ✓

(ii)

$N_{10} = 9.50 \times 10^{15} \times e^{-0.00122 \times 60}$ seen ✓ $N_{10} = 4.57 \times 10^{15}$ ✓

So number of aluminium-27 nuclei = $(9.50 - 4.57) \times 10^{15} = 4.9(3) \times 10^{15}$ ✓

(iii)

Total energy released = ans (c)(ii) $\times 2.62 \times 10^6 \times 1.6 \times 10^{-19}$ «= 2100 J» ✓

« $\frac{2100}{600} = \text{» } 3.4 - 3.5$ «W» ✓

(d)

ALTERNATIVE 1 «the process is adiabatic so» $\Delta Q = 0$ ✓ **ALTERNATIVE 2**

The compression is reversible «so $\Delta S = 0$ » ✓ **OWTTE**

(e)

area under curve AB is less than area under curve BC ✓

Do **not** allow **ECF** from part (b)

(f)

«W = 0 so» $Q = \Delta U$ ✓ « $\Delta U = \frac{3}{2} \times 53.2 \times R \times (351 - 27)$ so » $\Delta U = 2.15 \times 10^5$ «J» ✓

Award [2] for BCA

(a)

$$0.180 + 0.200 \times 0.60^2 \ll = 0.252 \text{ kg m}^2 \gg \checkmark$$

(b)

$$\text{angular speed of particle} = \ll 12/0.6 = \gg 20 \text{ rad s}^{-1} \gg \quad \textbf{OR}$$

$$\text{angular momentum of particle} \ll 0.200 \times 12.0 \times 0.60 \gg = 1.44 \text{ Js} \gg \checkmark$$

$$\ll \text{angular momentum of rod-particle system} \gg 0.252 \omega \gg$$

$$\text{equating } \omega = \ll \frac{1.44}{0.252} \gg = 5.71 \text{ rad s}^{-1} \checkmark$$

For **MP2**, working or answer to at least 3 SF should be seen.

(c)

$$\frac{1}{2} \times 0.200 \times 12.0^2 - \frac{1}{2} (0.252) \times 5.71^2 \checkmark \quad 10.3 \text{ J} \checkmark$$

Award [1] for answer 11.5 J that neglects moment of inertia of particle but do not penalize this omission in (d)(i).

(i)

$$\alpha = \frac{0.152}{0.252} = 0.603 \text{ rads}^{-2} \checkmark \quad \text{Accept negative values.}$$

(ii)

$$\theta = \frac{5.71^2}{2 \times 0.603} = 27.0 \text{ rad} \checkmark \quad N = \frac{27.0}{2 \times \pi} = 4.3 \checkmark$$

(e)

the rod will rotate «about centre of mass» $\checkmark$

«centre of mass» will move along straight line

«parallel to the particle's initial velocity» $\checkmark$

For **MP2**, mention of translational motion is not enough.

23M.2.SL.TZ1.8

(a)

$$0.180 + 0.200 \times 0.60^2 \ll = 0.252 \text{ kg m}^2 \gg \checkmark$$

(b)

$$\text{angular speed of particle} = \ll 12/0.6 = \gg 20 \text{ rad s}^{-1} \gg \quad \textbf{OR}$$

angular momentum of particle « $0.200 \times 12.0 \times 0.60$ » = 1.44 «Js» ✓

«angular momentum of rod-particle system 0.252 ω »

equating $\omega = \left\langle \frac{1.44}{0.252} \right\rangle = 5.71 \text{ rad s}^{-1}$ ✓

For **MP2**, working or answer to at least 3 SF should be seen.

(c)

$$\frac{1}{2} \times 0.200 \times 12.0^2 - \frac{1}{2} (0.252) \times 5.71^2 \checkmark \quad 10.3 \text{ J } \checkmark$$

Award [1] for answer 11.5 J that neglects moment of inertia of particle but do not penalize this omission in **(d)(i)**.

(i)

$$\alpha = \frac{0.152}{0.252} = 0.603 \text{ rads}^{-2} \checkmark \quad \text{Accept negative values.}$$

(ii)

$$\theta = \frac{5.71^2}{2 \times 0.603} = 27.0 \text{ rad } \checkmark \quad N = \frac{27.0}{2 \times \pi} = 4.3 \checkmark$$

(e)

the rod will rotate «about centre of mass» ✓

«centre of mass» will move along straight line

«parallel to the particle's initial velocity» ✓

For **MP2**, mention of translational motion is not enough.

23M.2.SL.TZ2.5

(a)

«An atom with» the same number of protons **AND** different numbers of neutrons
OR

Same chemical properties **AND** different physical properties ✓

Do **not** allow just atomic number and mass number

(i)

3 ✓

(ii)

2 ✓

(e)

Alphas have double charge «and so are better ionisers» ✓

alphas have more mass and therefore slower «for same energy» ✓

so longer time/more likely to interact with the «atomic» electrons/atoms «and therefore better ionisers» ✓

Accept reverse argument in terms of betas travelling faster.

EXE.2.HL.TZ0.10

(a)

Attempt to find area of triangle ✓ $\frac{1}{2} \times 8.0 \times 180 = 720 \text{ «kg m}^2 \text{ s}^{-1}\text{»}$ ✓

(a)

Attempt to find area of triangle ✓ $\frac{1}{2} \times 8.0 \times 180 = 720 \text{ «kg m}^2 \text{ s}^{-1}\text{»}$ ✓

(b)

Use of $\Delta L = \Delta I \omega$ ✓ 2.6 kg m^2 ✓

(c)

Correct conversions to a consistent set of ω units ✓

$\frac{1}{2} \times L \times \omega_f^2 - \omega_i^2$ or correct substitution seen ✓ 113 kJ ✓

EXE.2.HL.TZ0.15

(i)

$$\ll \sqrt{\frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{6.70 \times 10^6}} = \gg 7.71 \times 10^3 \text{ «m s}^{-1}\text{»} \gg \checkmark$$

(ii)

$$\ll \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{6.70 \times 10^6}} = \gg 1.09 \times 10^4 \text{ «m s}^{-1}\text{»} \gg \checkmark$$

(i)

Negative ✓

(ii)

EXE.2.HL.TZ0.16

(a)

$$v_{\text{esc}} = \sqrt{\frac{2GM}{r}} \text{ AND } g = \frac{GM}{r^2} \text{ seen } \checkmark \quad v_{\text{esc}} = \sqrt{\frac{2gr^2}{r}} \checkmark \quad \text{Leading to } v_{\text{esc}} = \sqrt{2gr} \checkmark$$

$$1.2 \text{ km s}^{-1} \checkmark$$

(a)

$$v_{\text{esc}} = \sqrt{\frac{2GM}{r}} \text{ AND } g = \frac{GM}{r^2} \text{ seen } \checkmark \quad v_{\text{esc}} = \sqrt{\frac{2gr^2}{r}} \checkmark \quad \text{Leading to } v_{\text{esc}} = \sqrt{2gr} \checkmark$$

$$1.2 \text{ km s}^{-1} \checkmark$$

(b)

Object at equator has the maximum linear/tangential speed possible ✓

It therefore has maximum kinetic energy before takeoff (and this is not required from the fuel) ✓

Idea that the object is already moving in direction of planet before takeoff ✓

EXE.2.HL.TZ0.17

(a)

$$\frac{1.24 \times 10^{-6}}{7.80 \times 10^{-11}} \ll 15.9 \text{ keV} \gg \checkmark$$

(a)

$$\frac{1.24 \times 10^{-6}}{7.80 \times 10^{-11}} \ll 15.9 \text{ keV} \gg \checkmark$$

(b)

$$\text{Energy of incident photon} = \ll 15.9 \times 10^3 + 500 = \gg 16.4 \times 10^3 \text{ «eV» } \checkmark$$

$$\text{Wavelength of incident photon} = \ll \frac{1.24 \times 10^{-6}}{16.4 \times 10^3} = \gg 7.56 \times 10^{-11} \text{ «m» } \checkmark$$

(c)

The wavelength of the X-rays changes ✓

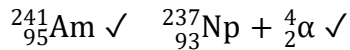
According to the wave model, the wavelength of the incident and scattered X-rays should be the same ✓

(d)

$$7.80 - 7.56 \times 10^{-11} = \frac{1.24 \times 10^{-6}}{0.511 \times 10^6} 1 - \cos \theta \quad \theta = 89^\circ \quad \checkmark$$

SPM.2.SL.TZ0.4

(a)



(c)

Each alpha gives rise to $\frac{5.5 \times 10^6}{15} = 3.67 \times 10^5$ ion pairs $\checkmark$

So $\frac{3.67 \times 10^5 \times 42000}{3} = 5.13 \times 10^9$ ion pairs per second $\checkmark$

current = $1.6 \times 10^{-19} \times 5.13 \times 10^9 = 0.82 \times 10^{-9}$ «A» $\checkmark$

SPM.2.HL.TZ0.9

(a.i)

equally spaced arrows «by eye» all pointing down $\checkmark$

edge effects also shown with arrows $\checkmark$

(i)

equally spaced arrows «by eye» all pointing down $\checkmark$

edge effects also shown with arrows $\checkmark$

(ii)

Weight vertically down **AND** electric force vertically up $\checkmark$

Of equal length «by eye» $\checkmark$

(iii)

Mass of drop is $\rho_0 V$ $\checkmark$ $qE = \rho_0 Vg$ $\checkmark$ «hence answer»

MP1 must be shown implicitly for credit.